

```
from google.colab import drive
drive.mount('/content/drive')
```

```
-----
-----
MessageError                                Traceback (most recent call
last)
```

```
/tmp/ipykernel_656/1408506528.py in <cell line: 0>()
```

```
      1 from google.colab import drive
----> 2 drive.mount('/content/drive')
```

```
/usr/local/lib/python3.12/dist-packages/google/colab/drive.py in
mount(mountpoint, force_remount, timeout_ms, readonly)
    95 def mount(mountpoint, force_remount=False, timeout_ms=120000,
readonly=False):
```

```
    96     """Mount your Google Drive at the specified mountpoint
path."""
```

```
---> 97     return _mount(
    98         mountpoint,
    99         force_remount=force_remount,
```

```
/usr/local/lib/python3.12/dist-packages/google/colab/drive.py in
_mount(mountpoint, force_remount, timeout_ms, ephemeral, readonly)
   132 )
```

```
   133 if ephemeral:
--> 134     _message.blocking_request(
   135         'request_auth',
   136         request={'authType': 'dfs_ephemeral'},
```

```
/usr/local/lib/python3.12/dist-packages/google/colab/_message.py in
blocking_request(request_type, request, timeout_sec, parent)
   174     request_type, request, parent=parent, expect_reply=True
   175 )
--> 176 return read_reply_from_input(request_id, timeout_sec)
```

```
/usr/local/lib/python3.12/dist-packages/google/colab/_message.py in
read_reply_from_input(message_id, timeout_sec)
```

```
   101 ):
   102     if 'error' in reply:
--> 103         raise MessageError(reply['error'])
   104     return reply.get('data', None)
   105
```

```
MessageError: Error: credential propagation was unsuccessful
```

```
# imports & config
import os
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```

from PIL import Image
from sklearn.model_selection import train_test_split
from sklearn.utils.class_weight import compute_class_weight
from sklearn.metrics import classification_report, confusion_matrix,
roc_curve, auc, accuracy_score, precision_score, recall_score,
f1_score
from sklearn.preprocessing import label_binarize
import tensorflow as tf
from tensorflow.keras import layers, optimizers, regularizers
from tensorflow.keras.applications import EfficientNetB0

DATASET_DIR      = "/content/drive/MyDrive/FINAL DATASET"
CLASSES          = ["CHAPPED", "HEALTHY", "MEDICAL"]
IMG_SIZE         = (224, 224)
BATCH_SIZE       = 16
SEED             = 12
EPOCHS_HEAD      = 60
EPOCHS_FINETUNE  = 120
UNFREEZE_LAYERS  = 20
DROPOUT_RATE     = 0.3
DENSE_UNITS      = 128
LR_HEAD          = 1e-4
LR_FINETUNE      = 1e-5

tf.random.set_seed(SEED)
np.random.seed(SEED)

# load images
images, labels = [], []
for label_idx, class_name in enumerate(CLASSES):
    class_dir = os.path.join(DATASET_DIR, class_name)
    files = sorted(f for f in os.listdir(class_dir) if
f.lower().endswith((".jpg", ".jpeg", ".png")))
    print(f"{class_name}: {len(files)} images")
    for fname in files:
        img_path = os.path.join(class_dir, fname)
        try:
            img = Image.open(img_path).convert("RGB")
            w, h = img.size
            min_dim = min(w, h)
            img = img.crop(((w-min_dim)//2, (h-min_dim)//2,
(w+min_dim)//2, (h+min_dim)//2))
            img = img.resize(IMG_SIZE, Image.LANCZOS)
            images.append(np.array(img, dtype=np.float32))
            labels.append(label_idx)
        except Exception as e:
            print(f"Skipping {img_path}: {e}")

X = np.array(images)

```

```

y = np.array(labels)
print(f"Total: {len(X)} images")

CHAPPED: 150 images
HEALTHY: 150 images
MEDICAL: 150 images
Total: 450 images

# split & class weights
X_train, X_temp, y_train, y_temp = train_test_split(X, y,
test_size=0.20, stratify=y, random_state=SEED)
X_val, X_test, y_val, y_test = train_test_split(X_temp, y_temp,
test_size=0.50, stratify=y_temp, random_state=SEED)
print(f"Train: {len(X_train)} | Val: {len(X_val)} | Test:
{len(X_test)}")

class_weights_array = compute_class_weight(class_weight="balanced",
classes=np.unique(y_train), y=y_train)
class_weight_dict = dict(enumerate(class_weights_array))

Train: 360 | Val: 45 | Test: 45

# augmentation & datasets
augment_layer = tf.keras.Sequential([
    layers.RandomFlip("horizontal"),
    layers.RandomRotation(0.15),
    layers.RandomZoom(0.15),
    layers.RandomBrightness(0.25),
    layers.RandomContrast(0.25),
])

def make_dataset(X, y, augment=False):
    ds = tf.data.Dataset.from_tensor_slices((X, y))
    if augment:
        ds = ds.map(lambda img, lbl: (augment_layer(img,
training=True), lbl), num_parallel_calls=tf.data.AUTOTUNE)
    return ds.batch(BATCH_SIZE).prefetch(tf.data.AUTOTUNE)

train_ds = make_dataset(X_train, y_train, augment=True)
val_ds = make_dataset(X_val, y_val)
test_ds = make_dataset(X_test, y_test)

# model definitions

# --- efficientnet ---
base = EfficientNetB0(input_shape=(*IMG_SIZE, 3), include_top=False,
weights="imagenet")
base.trainable = False

eff_inputs = tf.keras.Input(shape=(*IMG_SIZE, 3))
eff_x = base(eff_inputs, training=False)

```

```

eff_x      = layers.GlobalAveragePooling2D()(eff_x)
eff_x      = layers.BatchNormalization()(eff_x)
eff_x      = layers.Dense(DENSE_UNITS, activation="relu",
kernel_regularizer=regularizers.l2(1e-4))(eff_x)
eff_x      = layers.Dropout(DROPOUT_RATE)(eff_x)
eff_outputs = layers.Dense(len(CLASSES), activation="softmax")(eff_x)
eff_model  = tf.keras.Model(eff_inputs, eff_outputs)
eff_model.summary()

```

*# --- vanilla cnn ---*

```

cnn_inputs = tf.keras.Input(shape=(*IMG_SIZE, 3))
cnn_x = layers.Rescaling(1.0 / 255)(cnn_inputs)

cnn_x = layers.Conv2D(32, 3, padding="same",
kernel_regularizer=regularizers.l2(1e-4))(cnn_x)
cnn_x = layers.BatchNormalization()(cnn_x); cnn_x =
layers.Activation("relu")(cnn_x)
cnn_x = layers.Conv2D(32, 3, padding="same",
kernel_regularizer=regularizers.l2(1e-4))(cnn_x)
cnn_x = layers.BatchNormalization()(cnn_x); cnn_x =
layers.Activation("relu")(cnn_x)
cnn_x = layers.MaxPooling2D(2)(cnn_x)

cnn_x = layers.Conv2D(64, 3, padding="same",
kernel_regularizer=regularizers.l2(1e-4))(cnn_x)
cnn_x = layers.BatchNormalization()(cnn_x); cnn_x =
layers.Activation("relu")(cnn_x)
cnn_x = layers.Conv2D(64, 3, padding="same",
kernel_regularizer=regularizers.l2(1e-4))(cnn_x)
cnn_x = layers.BatchNormalization()(cnn_x); cnn_x =
layers.Activation("relu")(cnn_x)
cnn_x = layers.MaxPooling2D(2)(cnn_x)

cnn_x = layers.Conv2D(128, 3, padding="same",
kernel_regularizer=regularizers.l2(1e-4))(cnn_x)
cnn_x = layers.BatchNormalization()(cnn_x); cnn_x =
layers.Activation("relu")(cnn_x)
cnn_x = layers.Conv2D(128, 3, padding="same",
kernel_regularizer=regularizers.l2(1e-4))(cnn_x)
cnn_x = layers.BatchNormalization()(cnn_x); cnn_x =
layers.Activation("relu")(cnn_x)
cnn_x = layers.MaxPooling2D(2)(cnn_x)

cnn_x = layers.Conv2D(256, 3, padding="same",
kernel_regularizer=regularizers.l2(1e-4))(cnn_x)
cnn_x = layers.BatchNormalization()(cnn_x); cnn_x =
layers.Activation("relu")(cnn_x)
cnn_x = layers.Conv2D(256, 3, padding="same",
kernel_regularizer=regularizers.l2(1e-4))(cnn_x)
cnn_x = layers.BatchNormalization()(cnn_x); cnn_x =

```

```

layers.Activation("relu")(cnn_x)
cnn_x = layers.MaxPooling2D(2)(cnn_x)

cnn_x = layers.GlobalAveragePooling2D()(cnn_x)
cnn_x = layers.BatchNormalization()(cnn_x)
cnn_x = layers.Dense(DENSE_UNITS, activation="relu",
kernel_regularizer=regularizers.l2(1e-4))(cnn_x)
cnn_x = layers.Dropout(DROPOUT_RATE)(cnn_x)
cnn_outputs = layers.Dense(len(CLASSES), activation="softmax")(cnn_x)
cnn_model = tf.keras.Model(cnn_inputs, cnn_outputs)
cnn_model.summary()

```

Downloading data from [https://storage.googleapis.com/keras-applications/efficientnetb0\\_notop.h5](https://storage.googleapis.com/keras-applications/efficientnetb0_notop.h5)  
 16705208/16705208 \_\_\_\_\_ 0s 0us/step

Model: "functional\_1"

| Layer (type)<br>Param #                                   | Output Shape        |  |
|-----------------------------------------------------------|---------------------|--|
| input_layer_2 (InputLayer)<br>0                           | (None, 224, 224, 3) |  |
| efficientnetb0 (Functional)<br>4,049,571                  | (None, 7, 7, 1280)  |  |
| global_average_pooling2d<br>0<br>(GlobalAveragePooling2D) | (None, 1280)        |  |
| batch_normalization<br>5,120<br>(BatchNormalization)      | (None, 1280)        |  |
| dense (Dense)<br>163,968                                  | (None, 128)         |  |
| dropout (Dropout)<br>0                                    | (None, 128)         |  |

|                 |           |  |
|-----------------|-----------|--|
|                 |           |  |
| dense_1 (Dense) | (None, 3) |  |
| 387             |           |  |
|                 |           |  |

Total params: 4,219,046 (16.09 MB)

Trainable params: 166,915 (652.01 KB)

Non-trainable params: 4,052,131 (15.46 MB)

Model: "functional\_2"

| Layer (type)<br>Param #    | Output Shape         |  |
|----------------------------|----------------------|--|
| input_layer_3 (InputLayer) | (None, 224, 224, 3)  |  |
| 0                          |                      |  |
| rescaling_2 (Rescaling)    | (None, 224, 224, 3)  |  |
| 0                          |                      |  |
| conv2d (Conv2D)            | (None, 224, 224, 32) |  |
| 896                        |                      |  |
| batch_normalization_1      | (None, 224, 224, 32) |  |
| 128                        |                      |  |
| (BatchNormalization)       |                      |  |
| activation (Activation)    | (None, 224, 224, 32) |  |
| 0                          |                      |  |
| conv2d_1 (Conv2D)          | (None, 224, 224, 32) |  |
| 9,248                      |                      |  |
| batch_normalization_2      | (None, 224, 224, 32) |  |
| 128                        |                      |  |
| (BatchNormalization)       |                      |  |

|        |                                            |                      |
|--------|--------------------------------------------|----------------------|
|        |                                            |                      |
| 0      | activation_1 (Activation)                  | (None, 224, 224, 32) |
| 0      | max_pooling2d (MaxPooling2D)               | (None, 112, 112, 32) |
| 18,496 | conv2d_2 (Conv2D)                          | (None, 112, 112, 64) |
| 256    | batch_normalization_3 (BatchNormalization) | (None, 112, 112, 64) |
| 0      | activation_2 (Activation)                  | (None, 112, 112, 64) |
| 36,928 | conv2d_3 (Conv2D)                          | (None, 112, 112, 64) |
| 256    | batch_normalization_4 (BatchNormalization) | (None, 112, 112, 64) |
| 0      | activation_3 (Activation)                  | (None, 112, 112, 64) |
| 0      | max_pooling2d_1 (MaxPooling2D)             | (None, 56, 56, 64)   |
| 73,856 | conv2d_4 (Conv2D)                          | (None, 56, 56, 128)  |
| 512    | batch_normalization_5                      | (None, 56, 56, 128)  |

|         |                                |                     |  |
|---------|--------------------------------|---------------------|--|
|         | (BatchNormalization)           |                     |  |
|         |                                |                     |  |
|         |                                |                     |  |
| 0       | activation_4 (Activation)      | (None, 56, 56, 128) |  |
|         |                                |                     |  |
|         |                                |                     |  |
| 147,584 | conv2d_5 (Conv2D)              | (None, 56, 56, 128) |  |
|         |                                |                     |  |
|         |                                |                     |  |
| 512     | batch_normalization_6          | (None, 56, 56, 128) |  |
|         | (BatchNormalization)           |                     |  |
|         |                                |                     |  |
|         |                                |                     |  |
| 0       | activation_5 (Activation)      | (None, 56, 56, 128) |  |
|         |                                |                     |  |
|         |                                |                     |  |
| 0       | max_pooling2d_2 (MaxPooling2D) | (None, 28, 28, 128) |  |
|         |                                |                     |  |
|         |                                |                     |  |
| 295,168 | conv2d_6 (Conv2D)              | (None, 28, 28, 256) |  |
|         |                                |                     |  |
|         |                                |                     |  |
| 1,024   | batch_normalization_7          | (None, 28, 28, 256) |  |
|         | (BatchNormalization)           |                     |  |
|         |                                |                     |  |
|         |                                |                     |  |
| 0       | activation_6 (Activation)      | (None, 28, 28, 256) |  |
|         |                                |                     |  |
|         |                                |                     |  |
| 590,080 | conv2d_7 (Conv2D)              | (None, 28, 28, 256) |  |
|         |                                |                     |  |
|         |                                |                     |  |
| 1,024   | batch_normalization_8          | (None, 28, 28, 256) |  |
|         | (BatchNormalization)           |                     |  |
|         |                                |                     |  |
|         |                                |                     |  |



|        |                                                     |                     |  |
|--------|-----------------------------------------------------|---------------------|--|
| 0      | activation_7 (Activation)                           | (None, 28, 28, 256) |  |
| 0      | max_pooling2d_3 (MaxPooling2D)                      | (None, 14, 14, 256) |  |
| 0      | global_average_pooling2d_1 (GlobalAveragePooling2D) | (None, 256)         |  |
| 1,024  | batch_normalization_9 (BatchNormalization)          | (None, 256)         |  |
| 32,896 | dense_2 (Dense)                                     | (None, 128)         |  |
| 0      | dropout_1 (Dropout)                                 | (None, 128)         |  |
| 387    | dense_3 (Dense)                                     | (None, 3)           |  |

Total params: 1,210,403 (4.62 MB)

Trainable params: 1,207,971 (4.61 MB)

Non-trainable params: 2,432 (9.50 KB)

*# phase 1 – train both*

*# --- efficientnet ---*

```
eff_model.compile(optimizer=optimizers.Adam(LR_HEAD),
loss="sparse_categorical_crossentropy", metrics=["accuracy"])
eff_hist_1 = eff_model.fit(train_ds, validation_data=val_ds,
epochs=EPOCHS_HEAD, class_weight=class_weight_dict, verbose=1)
```

*# --- vanilla cnn ---*

```
cnn_model.compile(optimizer=optimizers.Adam(LR_HEAD),
```

```
loss="sparse_categorical_crossentropy", metrics=["accuracy"])
cnn_hist_1 = cnn_model.fit(train_ds, validation_data=val_ds,
epochs=EPOCHS_HEAD, class_weight=class_weight_dict, verbose=1)
```

Epoch 1/60

23/23 \_\_\_\_\_ 71s 2s/step - accuracy: 0.4250 - loss:  
1.3660 - val\_accuracy: 0.5778 - val\_loss: 0.9890

Epoch 2/60

23/23 \_\_\_\_\_ 4s 157ms/step - accuracy: 0.5694 - loss:  
1.0489 - val\_accuracy: 0.6000 - val\_loss: 0.9143

Epoch 3/60

23/23 \_\_\_\_\_ 4s 158ms/step - accuracy: 0.6139 - loss:  
0.9294 - val\_accuracy: 0.6444 - val\_loss: 0.8536

Epoch 4/60

23/23 \_\_\_\_\_ 6s 246ms/step - accuracy: 0.6389 - loss:  
0.8688 - val\_accuracy: 0.6667 - val\_loss: 0.8150

Epoch 5/60

23/23 \_\_\_\_\_ 4s 159ms/step - accuracy: 0.6444 - loss:  
0.8205 - val\_accuracy: 0.6889 - val\_loss: 0.7838

Epoch 6/60

23/23 \_\_\_\_\_ 4s 159ms/step - accuracy: 0.6806 - loss:  
0.7684 - val\_accuracy: 0.6889 - val\_loss: 0.7512

Epoch 7/60

23/23 \_\_\_\_\_ 7s 244ms/step - accuracy: 0.6889 - loss:  
0.7763 - val\_accuracy: 0.6889 - val\_loss: 0.7248

Epoch 8/60

23/23 \_\_\_\_\_ 4s 162ms/step - accuracy: 0.7056 - loss:  
0.7112 - val\_accuracy: 0.7333 - val\_loss: 0.6945

Epoch 9/60

23/23 \_\_\_\_\_ 5s 159ms/step - accuracy: 0.7639 - loss:  
0.6780 - val\_accuracy: 0.7111 - val\_loss: 0.6633

Epoch 10/60

23/23 \_\_\_\_\_ 6s 246ms/step - accuracy: 0.7472 - loss:  
0.6325 - val\_accuracy: 0.7778 - val\_loss: 0.6350

Epoch 11/60

23/23 \_\_\_\_\_ 4s 160ms/step - accuracy: 0.7500 - loss:  
0.6663 - val\_accuracy: 0.7111 - val\_loss: 0.6194

Epoch 12/60

23/23 \_\_\_\_\_ 4s 158ms/step - accuracy: 0.7944 - loss:  
0.5544 - val\_accuracy: 0.7556 - val\_loss: 0.6056

Epoch 13/60

23/23 \_\_\_\_\_ 5s 204ms/step - accuracy: 0.7806 - loss:  
0.6059 - val\_accuracy: 0.7333 - val\_loss: 0.5865

Epoch 14/60

23/23 \_\_\_\_\_ 5s 186ms/step - accuracy: 0.7500 - loss:  
0.5498 - val\_accuracy: 0.7778 - val\_loss: 0.5738

Epoch 15/60

23/23 \_\_\_\_\_ 4s 160ms/step - accuracy: 0.8028 - loss:  
0.5536 - val\_accuracy: 0.7556 - val\_loss: 0.5703

Epoch 16/60

23/23 \_\_\_\_\_ 4s 165ms/step - accuracy: 0.8028 - loss: 0.5219 - val\_accuracy: 0.7556 - val\_loss: 0.5551  
Epoch 17/60

23/23 \_\_\_\_\_ 5s 229ms/step - accuracy: 0.8222 - loss: 0.4998 - val\_accuracy: 0.7556 - val\_loss: 0.5446  
Epoch 18/60

23/23 \_\_\_\_\_ 4s 158ms/step - accuracy: 0.8500 - loss: 0.4590 - val\_accuracy: 0.7556 - val\_loss: 0.5419  
Epoch 19/60

23/23 \_\_\_\_\_ 4s 159ms/step - accuracy: 0.8250 - loss: 0.4787 - val\_accuracy: 0.7556 - val\_loss: 0.5494  
Epoch 20/60

23/23 \_\_\_\_\_ 5s 234ms/step - accuracy: 0.8222 - loss: 0.4621 - val\_accuracy: 0.8000 - val\_loss: 0.5434  
Epoch 21/60

23/23 \_\_\_\_\_ 4s 161ms/step - accuracy: 0.8444 - loss: 0.4121 - val\_accuracy: 0.8000 - val\_loss: 0.5270  
Epoch 22/60

23/23 \_\_\_\_\_ 5s 157ms/step - accuracy: 0.8139 - loss: 0.4616 - val\_accuracy: 0.8000 - val\_loss: 0.5263  
Epoch 23/60

23/23 \_\_\_\_\_ 5s 229ms/step - accuracy: 0.8500 - loss: 0.4488 - val\_accuracy: 0.8000 - val\_loss: 0.5289  
Epoch 24/60

23/23 \_\_\_\_\_ 4s 162ms/step - accuracy: 0.8500 - loss: 0.3838 - val\_accuracy: 0.7778 - val\_loss: 0.5259  
Epoch 25/60

23/23 \_\_\_\_\_ 4s 159ms/step - accuracy: 0.8472 - loss: 0.4088 - val\_accuracy: 0.7778 - val\_loss: 0.5177  
Epoch 26/60

23/23 \_\_\_\_\_ 4s 180ms/step - accuracy: 0.8389 - loss: 0.4068 - val\_accuracy: 0.7778 - val\_loss: 0.5129  
Epoch 27/60

23/23 \_\_\_\_\_ 5s 174ms/step - accuracy: 0.8528 - loss: 0.3813 - val\_accuracy: 0.7556 - val\_loss: 0.5222  
Epoch 28/60

23/23 \_\_\_\_\_ 4s 157ms/step - accuracy: 0.8389 - loss: 0.4418 - val\_accuracy: 0.7778 - val\_loss: 0.5296  
Epoch 29/60

23/23 \_\_\_\_\_ 6s 212ms/step - accuracy: 0.8806 - loss: 0.3574 - val\_accuracy: 0.7778 - val\_loss: 0.5233  
Epoch 30/60

23/23 \_\_\_\_\_ 4s 176ms/step - accuracy: 0.8750 - loss: 0.3758 - val\_accuracy: 0.7778 - val\_loss: 0.5265  
Epoch 31/60

23/23 \_\_\_\_\_ 4s 157ms/step - accuracy: 0.8556 - loss: 0.4019 - val\_accuracy: 0.7778 - val\_loss: 0.5100  
Epoch 32/60

23/23 \_\_\_\_\_ 4s 163ms/step - accuracy: 0.8500 - loss:

0.3890 - val\_accuracy: 0.7778 - val\_loss: 0.5100  
Epoch 33/60  
23/23 \_\_\_\_\_ 5s 228ms/step - accuracy: 0.8722 - loss:  
0.3496 - val\_accuracy: 0.7778 - val\_loss: 0.5150  
Epoch 34/60  
23/23 \_\_\_\_\_ 4s 161ms/step - accuracy: 0.8694 - loss:  
0.3508 - val\_accuracy: 0.8000 - val\_loss: 0.5119  
Epoch 35/60  
23/23 \_\_\_\_\_ 4s 161ms/step - accuracy: 0.8944 - loss:  
0.3145 - val\_accuracy: 0.8000 - val\_loss: 0.5015  
Epoch 36/60  
23/23 \_\_\_\_\_ 6s 247ms/step - accuracy: 0.8583 - loss:  
0.3552 - val\_accuracy: 0.7778 - val\_loss: 0.5065  
Epoch 37/60  
23/23 \_\_\_\_\_ 4s 164ms/step - accuracy: 0.8806 - loss:  
0.3546 - val\_accuracy: 0.7778 - val\_loss: 0.5038  
Epoch 38/60  
23/23 \_\_\_\_\_ 4s 159ms/step - accuracy: 0.8917 - loss:  
0.3303 - val\_accuracy: 0.7778 - val\_loss: 0.4876  
Epoch 39/60  
23/23 \_\_\_\_\_ 7s 243ms/step - accuracy: 0.8833 - loss:  
0.3404 - val\_accuracy: 0.8000 - val\_loss: 0.4854  
Epoch 40/60  
23/23 \_\_\_\_\_ 4s 157ms/step - accuracy: 0.8667 - loss:  
0.3324 - val\_accuracy: 0.7778 - val\_loss: 0.4842  
Epoch 41/60  
23/23 \_\_\_\_\_ 4s 162ms/step - accuracy: 0.8972 - loss:  
0.3155 - val\_accuracy: 0.8000 - val\_loss: 0.4820  
Epoch 42/60  
23/23 \_\_\_\_\_ 5s 208ms/step - accuracy: 0.9056 - loss:  
0.3146 - val\_accuracy: 0.8000 - val\_loss: 0.4776  
Epoch 43/60  
23/23 \_\_\_\_\_ 5s 195ms/step - accuracy: 0.8889 - loss:  
0.2981 - val\_accuracy: 0.8000 - val\_loss: 0.4769  
Epoch 44/60  
23/23 \_\_\_\_\_ 4s 171ms/step - accuracy: 0.9028 - loss:  
0.3023 - val\_accuracy: 0.8000 - val\_loss: 0.4896  
Epoch 45/60  
23/23 \_\_\_\_\_ 5s 197ms/step - accuracy: 0.9000 - loss:  
0.3047 - val\_accuracy: 0.8000 - val\_loss: 0.4944  
Epoch 46/60  
23/23 \_\_\_\_\_ 6s 233ms/step - accuracy: 0.8694 - loss:  
0.3275 - val\_accuracy: 0.8222 - val\_loss: 0.4897  
Epoch 47/60  
23/23 \_\_\_\_\_ 4s 170ms/step - accuracy: 0.8889 - loss:  
0.3152 - val\_accuracy: 0.8222 - val\_loss: 0.5074  
Epoch 48/60  
23/23 \_\_\_\_\_ 4s 175ms/step - accuracy: 0.9167 - loss:  
0.2838 - val\_accuracy: 0.8222 - val\_loss: 0.5173

Epoch 49/60  
23/23 \_\_\_\_\_ 6s 245ms/step - accuracy: 0.8944 - loss: 0.2958 - val\_accuracy: 0.8222 - val\_loss: 0.5110  
Epoch 50/60  
23/23 \_\_\_\_\_ 4s 165ms/step - accuracy: 0.9333 - loss: 0.2629 - val\_accuracy: 0.8222 - val\_loss: 0.5032  
Epoch 51/60  
23/23 \_\_\_\_\_ 4s 159ms/step - accuracy: 0.9139 - loss: 0.2699 - val\_accuracy: 0.8222 - val\_loss: 0.5065  
Epoch 52/60  
23/23 \_\_\_\_\_ 6s 239ms/step - accuracy: 0.9222 - loss: 0.2505 - val\_accuracy: 0.8222 - val\_loss: 0.5120  
Epoch 53/60  
23/23 \_\_\_\_\_ 4s 156ms/step - accuracy: 0.9139 - loss: 0.2712 - val\_accuracy: 0.8222 - val\_loss: 0.5137  
Epoch 54/60  
23/23 \_\_\_\_\_ 5s 158ms/step - accuracy: 0.8861 - loss: 0.3028 - val\_accuracy: 0.8000 - val\_loss: 0.5313  
Epoch 55/60  
23/23 \_\_\_\_\_ 5s 234ms/step - accuracy: 0.8944 - loss: 0.2890 - val\_accuracy: 0.8222 - val\_loss: 0.5357  
Epoch 56/60  
23/23 \_\_\_\_\_ 4s 162ms/step - accuracy: 0.9056 - loss: 0.2673 - val\_accuracy: 0.8222 - val\_loss: 0.5351  
Epoch 57/60  
23/23 \_\_\_\_\_ 4s 161ms/step - accuracy: 0.9250 - loss: 0.2521 - val\_accuracy: 0.8222 - val\_loss: 0.5353  
Epoch 58/60  
23/23 \_\_\_\_\_ 4s 192ms/step - accuracy: 0.9167 - loss: 0.2779 - val\_accuracy: 0.8222 - val\_loss: 0.5218  
Epoch 59/60  
23/23 \_\_\_\_\_ 5s 184ms/step - accuracy: 0.9333 - loss: 0.2149 - val\_accuracy: 0.8000 - val\_loss: 0.5229  
Epoch 60/60  
23/23 \_\_\_\_\_ 4s 158ms/step - accuracy: 0.9056 - loss: 0.2683 - val\_accuracy: 0.8222 - val\_loss: 0.5291  
Epoch 1/60  
23/23 \_\_\_\_\_ 44s 967ms/step - accuracy: 0.4611 - loss: 1.4336 - val\_accuracy: 0.3333 - val\_loss: 1.2099  
Epoch 2/60  
23/23 \_\_\_\_\_ 7s 293ms/step - accuracy: 0.5444 - loss: 1.1571 - val\_accuracy: 0.3333 - val\_loss: 1.2709  
Epoch 3/60  
23/23 \_\_\_\_\_ 5s 216ms/step - accuracy: 0.6000 - loss: 1.0934 - val\_accuracy: 0.3333 - val\_loss: 1.3653  
Epoch 4/60  
23/23 \_\_\_\_\_ 6s 256ms/step - accuracy: 0.5944 - loss: 1.0211 - val\_accuracy: 0.3333 - val\_loss: 1.5626  
Epoch 5/60

23/23 \_\_\_\_\_ 9s 221ms/step - accuracy: 0.6056 - loss: 0.9896 - val\_accuracy: 0.3333 - val\_loss: 1.7357  
Epoch 6/60  
23/23 \_\_\_\_\_ 7s 293ms/step - accuracy: 0.6194 - loss: 0.9502 - val\_accuracy: 0.3333 - val\_loss: 2.3143  
Epoch 7/60  
23/23 \_\_\_\_\_ 5s 223ms/step - accuracy: 0.6389 - loss: 0.9117 - val\_accuracy: 0.3333 - val\_loss: 2.2138  
Epoch 8/60  
23/23 \_\_\_\_\_ 6s 241ms/step - accuracy: 0.6250 - loss: 0.9425 - val\_accuracy: 0.3333 - val\_loss: 2.4904  
Epoch 9/60  
23/23 \_\_\_\_\_ 10s 220ms/step - accuracy: 0.6583 - loss: 0.8621 - val\_accuracy: 0.3333 - val\_loss: 2.7090  
Epoch 10/60  
23/23 \_\_\_\_\_ 6s 279ms/step - accuracy: 0.6833 - loss: 0.8242 - val\_accuracy: 0.3333 - val\_loss: 3.0343  
Epoch 11/60  
23/23 \_\_\_\_\_ 5s 216ms/step - accuracy: 0.6833 - loss: 0.8006 - val\_accuracy: 0.3333 - val\_loss: 2.7713  
Epoch 12/60  
23/23 \_\_\_\_\_ 5s 217ms/step - accuracy: 0.6778 - loss: 0.7979 - val\_accuracy: 0.3333 - val\_loss: 2.8712  
Epoch 13/60  
23/23 \_\_\_\_\_ 6s 268ms/step - accuracy: 0.7083 - loss: 0.7755 - val\_accuracy: 0.3333 - val\_loss: 3.1955  
Epoch 14/60  
23/23 \_\_\_\_\_ 5s 209ms/step - accuracy: 0.7111 - loss: 0.7760 - val\_accuracy: 0.3333 - val\_loss: 3.2608  
Epoch 15/60  
23/23 \_\_\_\_\_ 6s 265ms/step - accuracy: 0.6889 - loss: 0.8144 - val\_accuracy: 0.3778 - val\_loss: 2.3688  
Epoch 16/60  
23/23 \_\_\_\_\_ 5s 223ms/step - accuracy: 0.7306 - loss: 0.7748 - val\_accuracy: 0.4667 - val\_loss: 1.9643  
Epoch 17/60  
23/23 \_\_\_\_\_ 5s 215ms/step - accuracy: 0.7083 - loss: 0.7650 - val\_accuracy: 0.4667 - val\_loss: 1.6892  
Epoch 18/60  
23/23 \_\_\_\_\_ 6s 280ms/step - accuracy: 0.7306 - loss: 0.7087 - val\_accuracy: 0.4000 - val\_loss: 1.9440  
Epoch 19/60  
23/23 \_\_\_\_\_ 9s 224ms/step - accuracy: 0.7750 - loss: 0.6479 - val\_accuracy: 0.4889 - val\_loss: 1.3171  
Epoch 20/60  
23/23 \_\_\_\_\_ 6s 266ms/step - accuracy: 0.7472 - loss: 0.6611 - val\_accuracy: 0.5111 - val\_loss: 1.7969  
Epoch 21/60  
23/23 \_\_\_\_\_ 5s 210ms/step - accuracy: 0.7583 - loss:

0.6819 - val\_accuracy: 0.5778 - val\_loss: 1.2095  
Epoch 22/60  
23/23 \_\_\_\_\_ 6s 263ms/step - accuracy: 0.7722 - loss:  
0.6580 - val\_accuracy: 0.5556 - val\_loss: 1.1582  
Epoch 23/60  
23/23 \_\_\_\_\_ 9s 224ms/step - accuracy: 0.7944 - loss:  
0.6139 - val\_accuracy: 0.5333 - val\_loss: 1.5350  
Epoch 24/60  
23/23 \_\_\_\_\_ 7s 288ms/step - accuracy: 0.7917 - loss:  
0.6169 - val\_accuracy: 0.6444 - val\_loss: 1.3200  
Epoch 25/60  
23/23 \_\_\_\_\_ 5s 219ms/step - accuracy: 0.7778 - loss:  
0.6336 - val\_accuracy: 0.6222 - val\_loss: 1.0873  
Epoch 26/60  
23/23 \_\_\_\_\_ 6s 249ms/step - accuracy: 0.7889 - loss:  
0.6202 - val\_accuracy: 0.5778 - val\_loss: 1.2577  
Epoch 27/60  
23/23 \_\_\_\_\_ 6s 249ms/step - accuracy: 0.7694 - loss:  
0.6361 - val\_accuracy: 0.7778 - val\_loss: 0.7996  
Epoch 28/60  
23/23 \_\_\_\_\_ 5s 215ms/step - accuracy: 0.7861 - loss:  
0.6051 - val\_accuracy: 0.4444 - val\_loss: 1.7100  
Epoch 29/60  
23/23 \_\_\_\_\_ 7s 284ms/step - accuracy: 0.7778 - loss:  
0.5982 - val\_accuracy: 0.5556 - val\_loss: 1.1442  
Epoch 30/60  
23/23 \_\_\_\_\_ 5s 209ms/step - accuracy: 0.8000 - loss:  
0.5637 - val\_accuracy: 0.6444 - val\_loss: 1.0452  
Epoch 31/60  
23/23 \_\_\_\_\_ 5s 233ms/step - accuracy: 0.8083 - loss:  
0.5486 - val\_accuracy: 0.6444 - val\_loss: 1.1022  
Epoch 32/60  
23/23 \_\_\_\_\_ 6s 260ms/step - accuracy: 0.7917 - loss:  
0.5657 - val\_accuracy: 0.6444 - val\_loss: 0.9398  
Epoch 33/60  
23/23 \_\_\_\_\_ 10s 269ms/step - accuracy: 0.8250 - loss:  
0.5113 - val\_accuracy: 0.4667 - val\_loss: 1.6867  
Epoch 34/60  
23/23 \_\_\_\_\_ 6s 233ms/step - accuracy: 0.8417 - loss:  
0.5050 - val\_accuracy: 0.6667 - val\_loss: 0.8853  
Epoch 35/60  
23/23 \_\_\_\_\_ 5s 218ms/step - accuracy: 0.8111 - loss:  
0.5427 - val\_accuracy: 0.5778 - val\_loss: 1.0135  
Epoch 36/60  
23/23 \_\_\_\_\_ 7s 286ms/step - accuracy: 0.8306 - loss:  
0.5053 - val\_accuracy: 0.6889 - val\_loss: 0.8874  
Epoch 37/60  
23/23 \_\_\_\_\_ 9s 225ms/step - accuracy: 0.8694 - loss:  
0.4762 - val\_accuracy: 0.6667 - val\_loss: 1.0347

Epoch 38/60  
23/23 ————— 6s 270ms/step - accuracy: 0.8278 - loss: 0.5274 - val\_accuracy: 0.5778 - val\_loss: 1.5313  
Epoch 39/60  
23/23 ————— 5s 209ms/step - accuracy: 0.8361 - loss: 0.4951 - val\_accuracy: 0.5778 - val\_loss: 1.1361  
Epoch 40/60  
23/23 ————— 7s 279ms/step - accuracy: 0.8000 - loss: 0.5906 - val\_accuracy: 0.6222 - val\_loss: 1.0593  
Epoch 41/60  
23/23 ————— 5s 217ms/step - accuracy: 0.8667 - loss: 0.4836 - val\_accuracy: 0.6444 - val\_loss: 1.0159  
Epoch 42/60  
23/23 ————— 5s 219ms/step - accuracy: 0.8417 - loss: 0.4928 - val\_accuracy: 0.5111 - val\_loss: 2.0074  
Epoch 43/60  
23/23 ————— 7s 278ms/step - accuracy: 0.8722 - loss: 0.4418 - val\_accuracy: 0.7111 - val\_loss: 0.8860  
Epoch 44/60  
23/23 ————— 5s 215ms/step - accuracy: 0.8500 - loss: 0.4989 - val\_accuracy: 0.6444 - val\_loss: 1.1170  
Epoch 45/60  
23/23 ————— 6s 260ms/step - accuracy: 0.8750 - loss: 0.4261 - val\_accuracy: 0.6444 - val\_loss: 1.2829  
Epoch 46/60  
23/23 ————— 9s 219ms/step - accuracy: 0.8417 - loss: 0.4724 - val\_accuracy: 0.6000 - val\_loss: 1.2075  
Epoch 47/60  
23/23 ————— 6s 280ms/step - accuracy: 0.8806 - loss: 0.4065 - val\_accuracy: 0.6444 - val\_loss: 0.9957  
Epoch 48/60  
23/23 ————— 9s 228ms/step - accuracy: 0.8833 - loss: 0.4301 - val\_accuracy: 0.6000 - val\_loss: 1.0763  
Epoch 49/60  
23/23 ————— 7s 295ms/step - accuracy: 0.8917 - loss: 0.3959 - val\_accuracy: 0.5556 - val\_loss: 1.4081  
Epoch 50/60  
23/23 ————— 5s 227ms/step - accuracy: 0.8778 - loss: 0.4250 - val\_accuracy: 0.6444 - val\_loss: 0.9289  
Epoch 51/60  
23/23 ————— 6s 277ms/step - accuracy: 0.8861 - loss: 0.4177 - val\_accuracy: 0.7111 - val\_loss: 0.9985  
Epoch 52/60  
23/23 ————— 9s 227ms/step - accuracy: 0.8694 - loss: 0.4168 - val\_accuracy: 0.6889 - val\_loss: 0.9009  
Epoch 53/60  
23/23 ————— 7s 294ms/step - accuracy: 0.8806 - loss: 0.4137 - val\_accuracy: 0.6000 - val\_loss: 1.0423  
Epoch 54/60



```

23/23 _____ 5s 223ms/step - accuracy: 0.8500 - loss:
0.4361 - val_accuracy: 0.6667 - val_loss: 1.0365
Epoch 55/60
23/23 _____ 11s 270ms/step - accuracy: 0.8917 - loss:
0.3933 - val_accuracy: 0.6444 - val_loss: 0.9144
Epoch 56/60
23/23 _____ 5s 224ms/step - accuracy: 0.9000 - loss:
0.4104 - val_accuracy: 0.7111 - val_loss: 0.9593
Epoch 57/60
23/23 _____ 7s 293ms/step - accuracy: 0.9139 - loss:
0.3288 - val_accuracy: 0.6222 - val_loss: 1.2833
Epoch 58/60
23/23 _____ 9s 223ms/step - accuracy: 0.8889 - loss:
0.3952 - val_accuracy: 0.6222 - val_loss: 1.2898
Epoch 59/60
23/23 _____ 7s 290ms/step - accuracy: 0.9000 - loss:
0.4049 - val_accuracy: 0.7111 - val_loss: 0.9146
Epoch 60/60
23/23 _____ 5s 223ms/step - accuracy: 0.9000 - loss:
0.3689 - val_accuracy: 0.6667 - val_loss: 1.0208

```

*# phase 2 – fine-tune both*

*# --- efficientnet ---*

```

base.trainable = True
for layer in base.layers[:-UNFREEZE_LAYERS]:
    layer.trainable = False
eff_model.compile(optimizer=optimizers.Adam(LR_FINETUNE),
loss="sparse_categorical_crossentropy", metrics=["accuracy"])
eff_hist_2 = eff_model.fit(train_ds, validation_data=val_ds,
epochs=EPOCHS_FINETUNE, class_weight=class_weight_dict, verbose=1)

```

*# --- vanilla cnn ---*

```

cnn_model.compile(optimizer=optimizers.Adam(LR_FINETUNE),
loss="sparse_categorical_crossentropy", metrics=["accuracy"])
cnn_hist_2 = cnn_model.fit(train_ds, validation_data=val_ds,
epochs=EPOCHS_FINETUNE, class_weight=class_weight_dict, verbose=1)

```

```

Epoch 1/120
23/23 _____ 60s 1s/step - accuracy: 0.7222 - loss:
0.6939 - val_accuracy: 0.8000 - val_loss: 0.4842
Epoch 2/120
23/23 _____ 6s 251ms/step - accuracy: 0.7444 - loss:
0.6268 - val_accuracy: 0.8000 - val_loss: 0.4813
Epoch 3/120
23/23 _____ 4s 170ms/step - accuracy: 0.7806 - loss:
0.6189 - val_accuracy: 0.8000 - val_loss: 0.4974
Epoch 4/120
23/23 _____ 5s 179ms/step - accuracy: 0.7500 - loss:
0.6301 - val_accuracy: 0.8000 - val_loss: 0.5075

```

Epoch 5/120  
23/23 \_\_\_\_\_ 6s 203ms/step - accuracy: 0.7833 - loss: 0.5736 - val\_accuracy: 0.8222 - val\_loss: 0.5157  
Epoch 6/120  
23/23 \_\_\_\_\_ 4s 165ms/step - accuracy: 0.7833 - loss: 0.5610 - val\_accuracy: 0.8444 - val\_loss: 0.5232  
Epoch 7/120  
23/23 \_\_\_\_\_ 4s 181ms/step - accuracy: 0.8306 - loss: 0.5097 - val\_accuracy: 0.8222 - val\_loss: 0.5299  
Epoch 8/120  
23/23 \_\_\_\_\_ 5s 226ms/step - accuracy: 0.8056 - loss: 0.5062 - val\_accuracy: 0.7778 - val\_loss: 0.5331  
Epoch 9/120  
23/23 \_\_\_\_\_ 9s 185ms/step - accuracy: 0.7639 - loss: 0.6130 - val\_accuracy: 0.7778 - val\_loss: 0.5332  
Epoch 10/120  
23/23 \_\_\_\_\_ 5s 229ms/step - accuracy: 0.7972 - loss: 0.5382 - val\_accuracy: 0.7556 - val\_loss: 0.5478  
Epoch 11/120  
23/23 \_\_\_\_\_ 9s 180ms/step - accuracy: 0.8111 - loss: 0.4802 - val\_accuracy: 0.7556 - val\_loss: 0.5482  
Epoch 12/120  
23/23 \_\_\_\_\_ 6s 236ms/step - accuracy: 0.7778 - loss: 0.5669 - val\_accuracy: 0.7111 - val\_loss: 0.5501  
Epoch 13/120  
23/23 \_\_\_\_\_ 4s 167ms/step - accuracy: 0.8361 - loss: 0.4719 - val\_accuracy: 0.7111 - val\_loss: 0.5562  
Epoch 14/120  
23/23 \_\_\_\_\_ 6s 192ms/step - accuracy: 0.8111 - loss: 0.4711 - val\_accuracy: 0.7111 - val\_loss: 0.5621  
Epoch 15/120  
23/23 \_\_\_\_\_ 5s 219ms/step - accuracy: 0.8083 - loss: 0.4710 - val\_accuracy: 0.7111 - val\_loss: 0.5630  
Epoch 16/120  
23/23 \_\_\_\_\_ 4s 167ms/step - accuracy: 0.8694 - loss: 0.4133 - val\_accuracy: 0.7111 - val\_loss: 0.5611  
Epoch 17/120  
23/23 \_\_\_\_\_ 4s 168ms/step - accuracy: 0.8167 - loss: 0.4829 - val\_accuracy: 0.7111 - val\_loss: 0.5598  
Epoch 18/120  
23/23 \_\_\_\_\_ 6s 250ms/step - accuracy: 0.8361 - loss: 0.4574 - val\_accuracy: 0.7111 - val\_loss: 0.5658  
Epoch 19/120  
23/23 \_\_\_\_\_ 8s 165ms/step - accuracy: 0.8500 - loss: 0.4409 - val\_accuracy: 0.6889 - val\_loss: 0.5672  
Epoch 20/120  
23/23 \_\_\_\_\_ 6s 249ms/step - accuracy: 0.8472 - loss: 0.4279 - val\_accuracy: 0.7111 - val\_loss: 0.5616  
Epoch 21/120

23/23 \_\_\_\_\_ 8s 166ms/step - accuracy: 0.8389 - loss: 0.4561 - val\_accuracy: 0.7111 - val\_loss: 0.5626  
Epoch 22/120

23/23 \_\_\_\_\_ 6s 251ms/step - accuracy: 0.8306 - loss: 0.4351 - val\_accuracy: 0.7111 - val\_loss: 0.5639  
Epoch 23/120

23/23 \_\_\_\_\_ 4s 169ms/step - accuracy: 0.8306 - loss: 0.4723 - val\_accuracy: 0.7111 - val\_loss: 0.5712  
Epoch 24/120

23/23 \_\_\_\_\_ 4s 168ms/step - accuracy: 0.8556 - loss: 0.3793 - val\_accuracy: 0.7111 - val\_loss: 0.5732  
Epoch 25/120

23/23 \_\_\_\_\_ 5s 230ms/step - accuracy: 0.8611 - loss: 0.3755 - val\_accuracy: 0.6889 - val\_loss: 0.5734  
Epoch 26/120

23/23 \_\_\_\_\_ 9s 169ms/step - accuracy: 0.8667 - loss: 0.3414 - val\_accuracy: 0.7333 - val\_loss: 0.5681  
Epoch 27/120

23/23 \_\_\_\_\_ 7s 252ms/step - accuracy: 0.8528 - loss: 0.3938 - val\_accuracy: 0.7111 - val\_loss: 0.5706  
Epoch 28/120

23/23 \_\_\_\_\_ 8s 164ms/step - accuracy: 0.8361 - loss: 0.4034 - val\_accuracy: 0.7111 - val\_loss: 0.5652  
Epoch 29/120

23/23 \_\_\_\_\_ 6s 239ms/step - accuracy: 0.8583 - loss: 0.3965 - val\_accuracy: 0.7111 - val\_loss: 0.5656  
Epoch 30/120

23/23 \_\_\_\_\_ 4s 167ms/step - accuracy: 0.8750 - loss: 0.3521 - val\_accuracy: 0.6889 - val\_loss: 0.5668  
Epoch 31/120

23/23 \_\_\_\_\_ 4s 166ms/step - accuracy: 0.8583 - loss: 0.3597 - val\_accuracy: 0.7111 - val\_loss: 0.5640  
Epoch 32/120

23/23 \_\_\_\_\_ 7s 244ms/step - accuracy: 0.8528 - loss: 0.3841 - val\_accuracy: 0.7111 - val\_loss: 0.5583  
Epoch 33/120

23/23 \_\_\_\_\_ 4s 169ms/step - accuracy: 0.8778 - loss: 0.3581 - val\_accuracy: 0.7111 - val\_loss: 0.5532  
Epoch 34/120

23/23 \_\_\_\_\_ 4s 169ms/step - accuracy: 0.8611 - loss: 0.3876 - val\_accuracy: 0.7111 - val\_loss: 0.5440  
Epoch 35/120

23/23 \_\_\_\_\_ 5s 217ms/step - accuracy: 0.8833 - loss: 0.3419 - val\_accuracy: 0.7111 - val\_loss: 0.5358  
Epoch 36/120

23/23 \_\_\_\_\_ 5s 200ms/step - accuracy: 0.8889 - loss: 0.3190 - val\_accuracy: 0.7111 - val\_loss: 0.5368  
Epoch 37/120

23/23 \_\_\_\_\_ 4s 168ms/step - accuracy: 0.8639 - loss:

0.3825 - val\_accuracy: 0.7333 - val\_loss: 0.5360  
Epoch 38/120  
23/23 \_\_\_\_\_ 6s 223ms/step - accuracy: 0.9083 - loss:  
0.3033 - val\_accuracy: 0.7333 - val\_loss: 0.5315  
Epoch 39/120  
23/23 \_\_\_\_\_ 5s 190ms/step - accuracy: 0.8722 - loss:  
0.3474 - val\_accuracy: 0.7333 - val\_loss: 0.5282  
Epoch 40/120  
23/23 \_\_\_\_\_ 4s 167ms/step - accuracy: 0.8889 - loss:  
0.3405 - val\_accuracy: 0.7333 - val\_loss: 0.5279  
Epoch 41/120  
23/23 \_\_\_\_\_ 6s 229ms/step - accuracy: 0.8861 - loss:  
0.3163 - val\_accuracy: 0.7333 - val\_loss: 0.5302  
Epoch 42/120  
23/23 \_\_\_\_\_ 9s 166ms/step - accuracy: 0.8778 - loss:  
0.3275 - val\_accuracy: 0.7333 - val\_loss: 0.5396  
Epoch 43/120  
23/23 \_\_\_\_\_ 5s 235ms/step - accuracy: 0.9139 - loss:  
0.2906 - val\_accuracy: 0.7333 - val\_loss: 0.5373  
Epoch 44/120  
23/23 \_\_\_\_\_ 4s 178ms/step - accuracy: 0.8889 - loss:  
0.3107 - val\_accuracy: 0.7333 - val\_loss: 0.5366  
Epoch 45/120  
23/23 \_\_\_\_\_ 4s 169ms/step - accuracy: 0.8889 - loss:  
0.3149 - val\_accuracy: 0.7333 - val\_loss: 0.5341  
Epoch 46/120  
23/23 \_\_\_\_\_ 5s 208ms/step - accuracy: 0.8917 - loss:  
0.3244 - val\_accuracy: 0.7333 - val\_loss: 0.5333  
Epoch 47/120  
23/23 \_\_\_\_\_ 5s 198ms/step - accuracy: 0.9028 - loss:  
0.2854 - val\_accuracy: 0.7333 - val\_loss: 0.5307  
Epoch 48/120  
23/23 \_\_\_\_\_ 4s 168ms/step - accuracy: 0.8806 - loss:  
0.3161 - val\_accuracy: 0.7556 - val\_loss: 0.5330  
Epoch 49/120  
23/23 \_\_\_\_\_ 4s 179ms/step - accuracy: 0.9028 - loss:  
0.2908 - val\_accuracy: 0.7556 - val\_loss: 0.5383  
Epoch 50/120  
23/23 \_\_\_\_\_ 6s 197ms/step - accuracy: 0.8944 - loss:  
0.3126 - val\_accuracy: 0.7556 - val\_loss: 0.5371  
Epoch 51/120  
23/23 \_\_\_\_\_ 4s 169ms/step - accuracy: 0.8833 - loss:  
0.3344 - val\_accuracy: 0.7556 - val\_loss: 0.5423  
Epoch 52/120  
23/23 \_\_\_\_\_ 4s 190ms/step - accuracy: 0.9250 - loss:  
0.2573 - val\_accuracy: 0.7556 - val\_loss: 0.5432  
Epoch 53/120  
23/23 \_\_\_\_\_ 5s 198ms/step - accuracy: 0.8889 - loss:  
0.2896 - val\_accuracy: 0.7778 - val\_loss: 0.5405

Epoch 54/120  
23/23 \_\_\_\_\_ 4s 168ms/step - accuracy: 0.8944 - loss: 0.3001 - val\_accuracy: 0.7778 - val\_loss: 0.5345  
Epoch 55/120  
23/23 \_\_\_\_\_ 4s 177ms/step - accuracy: 0.9083 - loss: 0.3116 - val\_accuracy: 0.7556 - val\_loss: 0.5319  
Epoch 56/120  
23/23 \_\_\_\_\_ 6s 198ms/step - accuracy: 0.9167 - loss: 0.2491 - val\_accuracy: 0.7556 - val\_loss: 0.5260  
Epoch 57/120  
23/23 \_\_\_\_\_ 4s 166ms/step - accuracy: 0.9194 - loss: 0.2718 - val\_accuracy: 0.7556 - val\_loss: 0.5261  
Epoch 58/120  
23/23 \_\_\_\_\_ 4s 185ms/step - accuracy: 0.9056 - loss: 0.2828 - val\_accuracy: 0.7778 - val\_loss: 0.5270  
Epoch 59/120  
23/23 \_\_\_\_\_ 6s 199ms/step - accuracy: 0.9111 - loss: 0.2706 - val\_accuracy: 0.7778 - val\_loss: 0.5293  
Epoch 60/120  
23/23 \_\_\_\_\_ 4s 190ms/step - accuracy: 0.9083 - loss: 0.2705 - val\_accuracy: 0.7778 - val\_loss: 0.5294  
Epoch 61/120  
23/23 \_\_\_\_\_ 9s 389ms/step - accuracy: 0.8944 - loss: 0.2862 - val\_accuracy: 0.7778 - val\_loss: 0.5300  
Epoch 62/120  
23/23 \_\_\_\_\_ 4s 172ms/step - accuracy: 0.9194 - loss: 0.2526 - val\_accuracy: 0.7778 - val\_loss: 0.5211  
Epoch 63/120  
23/23 \_\_\_\_\_ 4s 168ms/step - accuracy: 0.9139 - loss: 0.2509 - val\_accuracy: 0.7778 - val\_loss: 0.5228  
Epoch 64/120  
23/23 \_\_\_\_\_ 7s 242ms/step - accuracy: 0.8889 - loss: 0.2725 - val\_accuracy: 0.7556 - val\_loss: 0.5221  
Epoch 65/120  
23/23 \_\_\_\_\_ 4s 168ms/step - accuracy: 0.9083 - loss: 0.2632 - val\_accuracy: 0.7556 - val\_loss: 0.5209  
Epoch 66/120  
23/23 \_\_\_\_\_ 4s 167ms/step - accuracy: 0.9111 - loss: 0.2878 - val\_accuracy: 0.7556 - val\_loss: 0.5251  
Epoch 67/120  
23/23 \_\_\_\_\_ 6s 262ms/step - accuracy: 0.9167 - loss: 0.2821 - val\_accuracy: 0.7556 - val\_loss: 0.5231  
Epoch 68/120  
23/23 \_\_\_\_\_ 4s 170ms/step - accuracy: 0.9361 - loss: 0.2435 - val\_accuracy: 0.7556 - val\_loss: 0.5268  
Epoch 69/120  
23/23 \_\_\_\_\_ 5s 174ms/step - accuracy: 0.9250 - loss: 0.2304 - val\_accuracy: 0.7556 - val\_loss: 0.5274  
Epoch 70/120

23/23 \_\_\_\_\_ 6s 257ms/step - accuracy: 0.9028 - loss: 0.2875 - val\_accuracy: 0.7556 - val\_loss: 0.5229  
Epoch 71/120

23/23 \_\_\_\_\_ 4s 173ms/step - accuracy: 0.8917 - loss: 0.2618 - val\_accuracy: 0.7556 - val\_loss: 0.5242  
Epoch 72/120

23/23 \_\_\_\_\_ 5s 169ms/step - accuracy: 0.9306 - loss: 0.2112 - val\_accuracy: 0.7556 - val\_loss: 0.5243  
Epoch 73/120

23/23 \_\_\_\_\_ 6s 257ms/step - accuracy: 0.9194 - loss: 0.2457 - val\_accuracy: 0.7556 - val\_loss: 0.5233  
Epoch 74/120

23/23 \_\_\_\_\_ 4s 166ms/step - accuracy: 0.9222 - loss: 0.2561 - val\_accuracy: 0.7556 - val\_loss: 0.5167  
Epoch 75/120

23/23 \_\_\_\_\_ 4s 168ms/step - accuracy: 0.9167 - loss: 0.2466 - val\_accuracy: 0.7556 - val\_loss: 0.5144  
Epoch 76/120

23/23 \_\_\_\_\_ 6s 242ms/step - accuracy: 0.9194 - loss: 0.2535 - val\_accuracy: 0.7556 - val\_loss: 0.5087  
Epoch 77/120

23/23 \_\_\_\_\_ 9s 169ms/step - accuracy: 0.9389 - loss: 0.2021 - val\_accuracy: 0.7556 - val\_loss: 0.5149  
Epoch 78/120

23/23 \_\_\_\_\_ 7s 259ms/step - accuracy: 0.9222 - loss: 0.2319 - val\_accuracy: 0.7556 - val\_loss: 0.5181  
Epoch 79/120

23/23 \_\_\_\_\_ 8s 166ms/step - accuracy: 0.9222 - loss: 0.2162 - val\_accuracy: 0.7556 - val\_loss: 0.5171  
Epoch 80/120

23/23 \_\_\_\_\_ 6s 258ms/step - accuracy: 0.9194 - loss: 0.2402 - val\_accuracy: 0.7556 - val\_loss: 0.5085  
Epoch 81/120

23/23 \_\_\_\_\_ 4s 169ms/step - accuracy: 0.9194 - loss: 0.2501 - val\_accuracy: 0.7556 - val\_loss: 0.5146  
Epoch 82/120

23/23 \_\_\_\_\_ 4s 170ms/step - accuracy: 0.9250 - loss: 0.2177 - val\_accuracy: 0.7556 - val\_loss: 0.5197  
Epoch 83/120

23/23 \_\_\_\_\_ 5s 227ms/step - accuracy: 0.9222 - loss: 0.2138 - val\_accuracy: 0.7556 - val\_loss: 0.5129  
Epoch 84/120

23/23 \_\_\_\_\_ 9s 165ms/step - accuracy: 0.9278 - loss: 0.2313 - val\_accuracy: 0.7556 - val\_loss: 0.5140  
Epoch 85/120

23/23 \_\_\_\_\_ 7s 252ms/step - accuracy: 0.9222 - loss: 0.2271 - val\_accuracy: 0.7556 - val\_loss: 0.5146  
Epoch 86/120

23/23 \_\_\_\_\_ 4s 166ms/step - accuracy: 0.9389 - loss:

0.1898 - val\_accuracy: 0.7556 - val\_loss: 0.5145  
Epoch 87/120  
23/23 \_\_\_\_\_ 4s 168ms/step - accuracy: 0.9222 - loss:  
0.2275 - val\_accuracy: 0.7556 - val\_loss: 0.5143  
Epoch 88/120  
23/23 \_\_\_\_\_ 5s 236ms/step - accuracy: 0.9417 - loss:  
0.2170 - val\_accuracy: 0.7556 - val\_loss: 0.5144  
Epoch 89/120  
23/23 \_\_\_\_\_ 4s 180ms/step - accuracy: 0.9472 - loss:  
0.1893 - val\_accuracy: 0.7556 - val\_loss: 0.5149  
Epoch 90/120  
23/23 \_\_\_\_\_ 4s 170ms/step - accuracy: 0.9389 - loss:  
0.1973 - val\_accuracy: 0.7556 - val\_loss: 0.4985  
Epoch 91/120  
23/23 \_\_\_\_\_ 7s 241ms/step - accuracy: 0.9444 - loss:  
0.1780 - val\_accuracy: 0.7556 - val\_loss: 0.4957  
Epoch 92/120  
23/23 \_\_\_\_\_ 4s 170ms/step - accuracy: 0.9278 - loss:  
0.2345 - val\_accuracy: 0.7333 - val\_loss: 0.4962  
Epoch 93/120  
23/23 \_\_\_\_\_ 4s 167ms/step - accuracy: 0.9222 - loss:  
0.2113 - val\_accuracy: 0.7556 - val\_loss: 0.4940  
Epoch 94/120  
23/23 \_\_\_\_\_ 6s 252ms/step - accuracy: 0.9500 - loss:  
0.1995 - val\_accuracy: 0.7556 - val\_loss: 0.4915  
Epoch 95/120  
23/23 \_\_\_\_\_ 4s 181ms/step - accuracy: 0.9389 - loss:  
0.1818 - val\_accuracy: 0.7556 - val\_loss: 0.4914  
Epoch 96/120  
23/23 \_\_\_\_\_ 4s 168ms/step - accuracy: 0.9389 - loss:  
0.2128 - val\_accuracy: 0.7556 - val\_loss: 0.4895  
Epoch 97/120  
23/23 \_\_\_\_\_ 7s 241ms/step - accuracy: 0.9250 - loss:  
0.2097 - val\_accuracy: 0.7556 - val\_loss: 0.4877  
Epoch 98/120  
23/23 \_\_\_\_\_ 9s 167ms/step - accuracy: 0.9528 - loss:  
0.1784 - val\_accuracy: 0.7556 - val\_loss: 0.4907  
Epoch 99/120  
23/23 \_\_\_\_\_ 5s 230ms/step - accuracy: 0.9444 - loss:  
0.1922 - val\_accuracy: 0.7556 - val\_loss: 0.4997  
Epoch 100/120  
23/23 \_\_\_\_\_ 9s 167ms/step - accuracy: 0.9194 - loss:  
0.2002 - val\_accuracy: 0.7556 - val\_loss: 0.5003  
Epoch 101/120  
23/23 \_\_\_\_\_ 5s 236ms/step - accuracy: 0.9472 - loss:  
0.1817 - val\_accuracy: 0.7556 - val\_loss: 0.5006  
Epoch 102/120  
23/23 \_\_\_\_\_ 9s 168ms/step - accuracy: 0.9361 - loss:  
0.2002 - val\_accuracy: 0.7556 - val\_loss: 0.5035

Epoch 103/120  
23/23 ————— 5s 232ms/step - accuracy: 0.9556 - loss: 0.1679 - val\_accuracy: 0.7556 - val\_loss: 0.5055  
Epoch 104/120  
23/23 ————— 9s 167ms/step - accuracy: 0.9333 - loss: 0.1892 - val\_accuracy: 0.7556 - val\_loss: 0.5110  
Epoch 105/120  
23/23 ————— 5s 227ms/step - accuracy: 0.9306 - loss: 0.2106 - val\_accuracy: 0.7556 - val\_loss: 0.5150  
Epoch 106/120  
23/23 ————— 4s 180ms/step - accuracy: 0.9472 - loss: 0.1836 - val\_accuracy: 0.7556 - val\_loss: 0.5169  
Epoch 107/120  
23/23 ————— 4s 170ms/step - accuracy: 0.9472 - loss: 0.1780 - val\_accuracy: 0.7556 - val\_loss: 0.5229  
Epoch 108/120  
23/23 ————— 5s 196ms/step - accuracy: 0.9417 - loss: 0.1811 - val\_accuracy: 0.7556 - val\_loss: 0.5356  
Epoch 109/120  
23/23 ————— 5s 216ms/step - accuracy: 0.9528 - loss: 0.1933 - val\_accuracy: 0.7556 - val\_loss: 0.5406  
Epoch 110/120  
23/23 ————— 4s 166ms/step - accuracy: 0.9667 - loss: 0.1523 - val\_accuracy: 0.7556 - val\_loss: 0.5427  
Epoch 111/120  
23/23 ————— 6s 202ms/step - accuracy: 0.9500 - loss: 0.1809 - val\_accuracy: 0.7556 - val\_loss: 0.5456  
Epoch 112/120  
23/23 ————— 5s 199ms/step - accuracy: 0.9611 - loss: 0.1769 - val\_accuracy: 0.7556 - val\_loss: 0.5427  
Epoch 113/120  
23/23 ————— 4s 168ms/step - accuracy: 0.9528 - loss: 0.1563 - val\_accuracy: 0.7556 - val\_loss: 0.5382  
Epoch 114/120  
23/23 ————— 6s 228ms/step - accuracy: 0.9361 - loss: 0.2049 - val\_accuracy: 0.7556 - val\_loss: 0.5332  
Epoch 115/120  
23/23 ————— 5s 188ms/step - accuracy: 0.9611 - loss: 0.1521 - val\_accuracy: 0.7556 - val\_loss: 0.5316  
Epoch 116/120  
23/23 ————— 5s 167ms/step - accuracy: 0.9194 - loss: 0.2210 - val\_accuracy: 0.7556 - val\_loss: 0.5314  
Epoch 117/120  
23/23 ————— 7s 252ms/step - accuracy: 0.9361 - loss: 0.1849 - val\_accuracy: 0.7556 - val\_loss: 0.5367  
Epoch 118/120  
23/23 ————— 4s 169ms/step - accuracy: 0.9556 - loss: 0.1686 - val\_accuracy: 0.7556 - val\_loss: 0.5336  
Epoch 119/120



23/23 ————— 4s 168ms/step - accuracy: 0.9417 - loss: 0.1751 - val\_accuracy: 0.7556 - val\_loss: 0.5300  
Epoch 120/120

23/23 ————— 7s 255ms/step - accuracy: 0.9306 - loss: 0.1820 - val\_accuracy: 0.7556 - val\_loss: 0.5245  
Epoch 1/120

23/23 ————— 29s 690ms/step - accuracy: 0.9250 - loss: 0.3147 - val\_accuracy: 0.7333 - val\_loss: 0.9114  
Epoch 2/120

23/23 ————— 23s 245ms/step - accuracy: 0.9444 - loss: 0.3008 - val\_accuracy: 0.7333 - val\_loss: 0.8819  
Epoch 3/120

23/23 ————— 6s 262ms/step - accuracy: 0.9583 - loss: 0.2741 - val\_accuracy: 0.7333 - val\_loss: 0.8889  
Epoch 4/120

23/23 ————— 5s 221ms/step - accuracy: 0.9500 - loss: 0.2946 - val\_accuracy: 0.7556 - val\_loss: 0.9340  
Epoch 5/120

23/23 ————— 7s 287ms/step - accuracy: 0.9361 - loss: 0.2859 - val\_accuracy: 0.7111 - val\_loss: 1.0236  
Epoch 6/120

23/23 ————— 5s 219ms/step - accuracy: 0.9417 - loss: 0.2774 - val\_accuracy: 0.7333 - val\_loss: 0.9226  
Epoch 7/120

23/23 ————— 5s 234ms/step - accuracy: 0.9556 - loss: 0.2671 - val\_accuracy: 0.7111 - val\_loss: 0.9204  
Epoch 8/120

23/23 ————— 6s 258ms/step - accuracy: 0.9306 - loss: 0.3040 - val\_accuracy: 0.6667 - val\_loss: 0.9117  
Epoch 9/120

23/23 ————— 5s 217ms/step - accuracy: 0.9583 - loss: 0.2524 - val\_accuracy: 0.7111 - val\_loss: 0.9889  
Epoch 10/120

23/23 ————— 7s 282ms/step - accuracy: 0.9333 - loss: 0.2879 - val\_accuracy: 0.7111 - val\_loss: 1.0126  
Epoch 11/120

23/23 ————— 5s 218ms/step - accuracy: 0.9500 - loss: 0.2598 - val\_accuracy: 0.6667 - val\_loss: 0.9144  
Epoch 12/120

23/23 ————— 5s 221ms/step - accuracy: 0.9750 - loss: 0.2285 - val\_accuracy: 0.6889 - val\_loss: 0.9014  
Epoch 13/120

23/23 ————— 10s 220ms/step - accuracy: 0.9056 - loss: 0.3247 - val\_accuracy: 0.7333 - val\_loss: 0.9204  
Epoch 14/120

23/23 ————— 6s 281ms/step - accuracy: 0.9500 - loss: 0.2569 - val\_accuracy: 0.7111 - val\_loss: 0.9433  
Epoch 15/120

23/23 ————— 5s 220ms/step - accuracy: 0.9583 - loss:

0.2472 - val\_accuracy: 0.6667 - val\_loss: 0.9207  
Epoch 16/120  
23/23 ————— 5s 219ms/step - accuracy: 0.9639 - loss: 0.2429 - val\_accuracy: 0.7111 - val\_loss: 0.9060  
Epoch 17/120  
23/23 ————— 6s 269ms/step - accuracy: 0.9583 - loss: 0.2576 - val\_accuracy: 0.7333 - val\_loss: 0.8626  
Epoch 18/120  
23/23 ————— 5s 216ms/step - accuracy: 0.9611 - loss: 0.2335 - val\_accuracy: 0.7111 - val\_loss: 0.8707  
Epoch 19/120  
23/23 ————— 6s 272ms/step - accuracy: 0.9611 - loss: 0.2485 - val\_accuracy: 0.6889 - val\_loss: 0.9296  
Epoch 20/120  
23/23 ————— 5s 219ms/step - accuracy: 0.9528 - loss: 0.2648 - val\_accuracy: 0.6889 - val\_loss: 0.8978  
Epoch 21/120  
23/23 ————— 5s 218ms/step - accuracy: 0.9667 - loss: 0.2419 - val\_accuracy: 0.7333 - val\_loss: 0.9392  
Epoch 22/120  
23/23 ————— 6s 281ms/step - accuracy: 0.9556 - loss: 0.2456 - val\_accuracy: 0.7333 - val\_loss: 0.9351  
Epoch 23/120  
23/23 ————— 5s 210ms/step - accuracy: 0.9667 - loss: 0.2300 - val\_accuracy: 0.7111 - val\_loss: 0.9432  
Epoch 24/120  
23/23 ————— 6s 252ms/step - accuracy: 0.9639 - loss: 0.2443 - val\_accuracy: 0.7333 - val\_loss: 0.9170  
Epoch 25/120  
23/23 ————— 6s 235ms/step - accuracy: 0.9722 - loss: 0.2357 - val\_accuracy: 0.7111 - val\_loss: 0.9276  
Epoch 26/120  
23/23 ————— 11s 284ms/step - accuracy: 0.9750 - loss: 0.2297 - val\_accuracy: 0.7111 - val\_loss: 0.9856  
Epoch 27/120  
23/23 ————— 5s 215ms/step - accuracy: 0.9444 - loss: 0.2520 - val\_accuracy: 0.7333 - val\_loss: 0.9242  
Epoch 28/120  
23/23 ————— 5s 231ms/step - accuracy: 0.9694 - loss: 0.2167 - val\_accuracy: 0.7111 - val\_loss: 0.9587  
Epoch 29/120  
23/23 ————— 10s 219ms/step - accuracy: 0.9722 - loss: 0.2259 - val\_accuracy: 0.6667 - val\_loss: 0.9685  
Epoch 30/120  
23/23 ————— 7s 282ms/step - accuracy: 0.9778 - loss: 0.2199 - val\_accuracy: 0.7111 - val\_loss: 0.9524  
Epoch 31/120  
23/23 ————— 5s 214ms/step - accuracy: 0.9722 - loss: 0.2172 - val\_accuracy: 0.7556 - val\_loss: 0.9732

Epoch 32/120  
23/23 ————— 5s 222ms/step - accuracy: 0.9472 - loss: 0.2482 - val\_accuracy: 0.7111 - val\_loss: 0.9818  
Epoch 33/120  
23/23 ————— 6s 268ms/step - accuracy: 0.9472 - loss: 0.2508 - val\_accuracy: 0.6667 - val\_loss: 1.0193  
Epoch 34/120  
23/23 ————— 5s 216ms/step - accuracy: 0.9611 - loss: 0.2399 - val\_accuracy: 0.7111 - val\_loss: 0.9395  
Epoch 35/120  
23/23 ————— 6s 273ms/step - accuracy: 0.9694 - loss: 0.2147 - val\_accuracy: 0.7111 - val\_loss: 1.0253  
Epoch 36/120  
23/23 ————— 5s 216ms/step - accuracy: 0.9639 - loss: 0.2249 - val\_accuracy: 0.7111 - val\_loss: 0.9796  
Epoch 37/120  
23/23 ————— 5s 213ms/step - accuracy: 0.9778 - loss: 0.2249 - val\_accuracy: 0.7556 - val\_loss: 0.9025  
Epoch 38/120  
23/23 ————— 6s 279ms/step - accuracy: 0.9806 - loss: 0.2106 - val\_accuracy: 0.7556 - val\_loss: 1.0296  
Epoch 39/120  
23/23 ————— 9s 233ms/step - accuracy: 0.9611 - loss: 0.2258 - val\_accuracy: 0.7333 - val\_loss: 1.0434  
Epoch 40/120  
23/23 ————— 6s 259ms/step - accuracy: 0.9889 - loss: 0.1976 - val\_accuracy: 0.6889 - val\_loss: 0.9286  
Epoch 41/120  
23/23 ————— 5s 213ms/step - accuracy: 0.9750 - loss: 0.2073 - val\_accuracy: 0.6667 - val\_loss: 0.9489  
Epoch 42/120  
23/23 ————— 7s 282ms/step - accuracy: 0.9750 - loss: 0.2129 - val\_accuracy: 0.7333 - val\_loss: 0.9514  
Epoch 43/120  
23/23 ————— 5s 211ms/step - accuracy: 0.9778 - loss: 0.1904 - val\_accuracy: 0.6889 - val\_loss: 0.8975  
Epoch 44/120  
23/23 ————— 5s 214ms/step - accuracy: 0.9778 - loss: 0.2169 - val\_accuracy: 0.7111 - val\_loss: 0.9857  
Epoch 45/120  
23/23 ————— 7s 277ms/step - accuracy: 0.9611 - loss: 0.2274 - val\_accuracy: 0.7111 - val\_loss: 0.9386  
Epoch 46/120  
23/23 ————— 5s 212ms/step - accuracy: 0.9667 - loss: 0.2194 - val\_accuracy: 0.7111 - val\_loss: 1.0016  
Epoch 47/120  
23/23 ————— 6s 256ms/step - accuracy: 0.9750 - loss: 0.2191 - val\_accuracy: 0.7111 - val\_loss: 1.0839  
Epoch 48/120  
23/23 ————— 9s 215ms/step - accuracy: 0.9639 - loss:

0.2315 - val\_accuracy: 0.7556 - val\_loss: 1.0386  
Epoch 49/120  
23/23 \_\_\_\_\_ 6s 280ms/step - accuracy: 0.9750 - loss:  
0.2156 - val\_accuracy: 0.7556 - val\_loss: 0.9996  
Epoch 50/120  
23/23 \_\_\_\_\_ 5s 208ms/step - accuracy: 0.9694 - loss:  
0.2059 - val\_accuracy: 0.6889 - val\_loss: 1.0576  
Epoch 51/120  
23/23 \_\_\_\_\_ 5s 223ms/step - accuracy: 0.9694 - loss:  
0.2036 - val\_accuracy: 0.6667 - val\_loss: 1.0500  
Epoch 52/120  
23/23 \_\_\_\_\_ 6s 262ms/step - accuracy: 0.9611 - loss:  
0.2292 - val\_accuracy: 0.6889 - val\_loss: 0.9582  
Epoch 53/120  
23/23 \_\_\_\_\_ 5s 214ms/step - accuracy: 0.9778 - loss:  
0.1947 - val\_accuracy: 0.6444 - val\_loss: 1.0277  
Epoch 54/120  
23/23 \_\_\_\_\_ 6s 257ms/step - accuracy: 0.9694 - loss:  
0.2177 - val\_accuracy: 0.6889 - val\_loss: 0.9782  
Epoch 55/120  
23/23 \_\_\_\_\_ 5s 226ms/step - accuracy: 0.9750 - loss:  
0.2010 - val\_accuracy: 0.7333 - val\_loss: 0.9562  
Epoch 56/120  
23/23 \_\_\_\_\_ 11s 280ms/step - accuracy: 0.9639 - loss:  
0.2223 - val\_accuracy: 0.7111 - val\_loss: 1.0140  
Epoch 57/120  
23/23 \_\_\_\_\_ 5s 210ms/step - accuracy: 0.9861 - loss:  
0.1963 - val\_accuracy: 0.7333 - val\_loss: 0.9430  
Epoch 58/120  
23/23 \_\_\_\_\_ 6s 260ms/step - accuracy: 0.9667 - loss:  
0.2067 - val\_accuracy: 0.7556 - val\_loss: 1.0684  
Epoch 59/120  
23/23 \_\_\_\_\_ 9s 214ms/step - accuracy: 0.9611 - loss:  
0.2277 - val\_accuracy: 0.6889 - val\_loss: 1.1464  
Epoch 60/120  
23/23 \_\_\_\_\_ 6s 277ms/step - accuracy: 0.9750 - loss:  
0.2164 - val\_accuracy: 0.7333 - val\_loss: 1.0234  
Epoch 61/120  
23/23 \_\_\_\_\_ 9s 217ms/step - accuracy: 0.9722 - loss:  
0.2161 - val\_accuracy: 0.7556 - val\_loss: 0.9563  
Epoch 62/120  
23/23 \_\_\_\_\_ 6s 277ms/step - accuracy: 0.9611 - loss:  
0.2271 - val\_accuracy: 0.7556 - val\_loss: 0.9547  
Epoch 63/120  
23/23 \_\_\_\_\_ 5s 211ms/step - accuracy: 0.9722 - loss:  
0.1956 - val\_accuracy: 0.7333 - val\_loss: 1.0003  
Epoch 64/120  
23/23 \_\_\_\_\_ 5s 222ms/step - accuracy: 0.9639 - loss:  
0.2085 - val\_accuracy: 0.7333 - val\_loss: 0.9923

Epoch 65/120  
23/23 \_\_\_\_\_ 10s 213ms/step - accuracy: 0.9667 - loss: 0.2119 - val\_accuracy: 0.7111 - val\_loss: 1.0029  
Epoch 66/120  
23/23 \_\_\_\_\_ 6s 277ms/step - accuracy: 0.9722 - loss: 0.2059 - val\_accuracy: 0.7111 - val\_loss: 0.9910  
Epoch 67/120  
23/23 \_\_\_\_\_ 5s 213ms/step - accuracy: 0.9861 - loss: 0.1748 - val\_accuracy: 0.6889 - val\_loss: 0.9663  
Epoch 68/120  
23/23 \_\_\_\_\_ 5s 212ms/step - accuracy: 0.9528 - loss: 0.2174 - val\_accuracy: 0.7333 - val\_loss: 0.9590  
Epoch 69/120  
23/23 \_\_\_\_\_ 6s 271ms/step - accuracy: 0.9750 - loss: 0.1976 - val\_accuracy: 0.7111 - val\_loss: 1.0645  
Epoch 70/120  
23/23 \_\_\_\_\_ 9s 236ms/step - accuracy: 0.9611 - loss: 0.2265 - val\_accuracy: 0.6444 - val\_loss: 1.1304  
Epoch 71/120  
23/23 \_\_\_\_\_ 10s 216ms/step - accuracy: 0.9639 - loss: 0.2347 - val\_accuracy: 0.7333 - val\_loss: 1.0300  
Epoch 72/120  
23/23 \_\_\_\_\_ 6s 275ms/step - accuracy: 0.9611 - loss: 0.1981 - val\_accuracy: 0.7556 - val\_loss: 1.0187  
Epoch 73/120  
23/23 \_\_\_\_\_ 5s 209ms/step - accuracy: 0.9694 - loss: 0.2047 - val\_accuracy: 0.7111 - val\_loss: 0.9439  
Epoch 74/120  
23/23 \_\_\_\_\_ 5s 213ms/step - accuracy: 0.9806 - loss: 0.1955 - val\_accuracy: 0.6889 - val\_loss: 1.0177  
Epoch 75/120  
23/23 \_\_\_\_\_ 6s 276ms/step - accuracy: 0.9667 - loss: 0.2094 - val\_accuracy: 0.6444 - val\_loss: 1.2564  
Epoch 76/120  
23/23 \_\_\_\_\_ 5s 213ms/step - accuracy: 0.9722 - loss: 0.2002 - val\_accuracy: 0.6000 - val\_loss: 1.3754  
Epoch 77/120  
23/23 \_\_\_\_\_ 6s 258ms/step - accuracy: 0.9750 - loss: 0.2052 - val\_accuracy: 0.7333 - val\_loss: 1.0463  
Epoch 78/120  
23/23 \_\_\_\_\_ 9s 212ms/step - accuracy: 0.9833 - loss: 0.1839 - val\_accuracy: 0.7556 - val\_loss: 1.1119  
Epoch 79/120  
23/23 \_\_\_\_\_ 6s 275ms/step - accuracy: 0.9722 - loss: 0.2024 - val\_accuracy: 0.6889 - val\_loss: 1.0881  
Epoch 80/120  
23/23 \_\_\_\_\_ 5s 207ms/step - accuracy: 0.9778 - loss: 0.1993 - val\_accuracy: 0.7333 - val\_loss: 1.1297  
Epoch 81/120

23/23 \_\_\_\_\_ 5s 215ms/step - accuracy: 0.9667 - loss: 0.2001 - val\_accuracy: 0.6889 - val\_loss: 1.1412  
Epoch 82/120

23/23 \_\_\_\_\_ 6s 266ms/step - accuracy: 0.9806 - loss: 0.1988 - val\_accuracy: 0.6889 - val\_loss: 1.1473  
Epoch 83/120

23/23 \_\_\_\_\_ 5s 211ms/step - accuracy: 0.9750 - loss: 0.1930 - val\_accuracy: 0.6889 - val\_loss: 1.1137  
Epoch 84/120

23/23 \_\_\_\_\_ 6s 259ms/step - accuracy: 0.9778 - loss: 0.1951 - val\_accuracy: 0.6889 - val\_loss: 1.0392  
Epoch 85/120

23/23 \_\_\_\_\_ 9s 213ms/step - accuracy: 0.9750 - loss: 0.2079 - val\_accuracy: 0.6667 - val\_loss: 1.0818  
Epoch 86/120

23/23 \_\_\_\_\_ 6s 279ms/step - accuracy: 0.9639 - loss: 0.2234 - val\_accuracy: 0.6889 - val\_loss: 1.0972  
Epoch 87/120

23/23 \_\_\_\_\_ 5s 214ms/step - accuracy: 0.9806 - loss: 0.1862 - val\_accuracy: 0.7111 - val\_loss: 1.1017  
Epoch 88/120

23/23 \_\_\_\_\_ 5s 228ms/step - accuracy: 0.9694 - loss: 0.2067 - val\_accuracy: 0.6889 - val\_loss: 1.1143  
Epoch 89/120

23/23 \_\_\_\_\_ 10s 216ms/step - accuracy: 0.9722 - loss: 0.1961 - val\_accuracy: 0.7333 - val\_loss: 1.0830  
Epoch 90/120

23/23 \_\_\_\_\_ 6s 277ms/step - accuracy: 0.9889 - loss: 0.1736 - val\_accuracy: 0.7333 - val\_loss: 1.1349  
Epoch 91/120

23/23 \_\_\_\_\_ 5s 217ms/step - accuracy: 0.9889 - loss: 0.1833 - val\_accuracy: 0.7111 - val\_loss: 1.1170  
Epoch 92/120

23/23 \_\_\_\_\_ 5s 214ms/step - accuracy: 0.9806 - loss: 0.1871 - val\_accuracy: 0.7333 - val\_loss: 1.0342  
Epoch 93/120

23/23 \_\_\_\_\_ 6s 265ms/step - accuracy: 0.9750 - loss: 0.1966 - val\_accuracy: 0.8000 - val\_loss: 0.9899  
Epoch 94/120

23/23 \_\_\_\_\_ 10s 248ms/step - accuracy: 0.9778 - loss: 0.1845 - val\_accuracy: 0.7333 - val\_loss: 0.9979  
Epoch 95/120

23/23 \_\_\_\_\_ 6s 237ms/step - accuracy: 0.9833 - loss: 0.1889 - val\_accuracy: 0.7111 - val\_loss: 1.0378  
Epoch 96/120

23/23 \_\_\_\_\_ 5s 211ms/step - accuracy: 0.9722 - loss: 0.1992 - val\_accuracy: 0.7111 - val\_loss: 1.0560  
Epoch 97/120

23/23 \_\_\_\_\_ 6s 277ms/step - accuracy: 0.9944 - loss:

0.1640 - val\_accuracy: 0.7333 - val\_loss: 1.0215  
Epoch 98/120  
23/23 ————— 5s 210ms/step - accuracy: 0.9778 - loss: 0.1804 - val\_accuracy: 0.7111 - val\_loss: 0.9757  
Epoch 99/120  
23/23 ————— 5s 216ms/step - accuracy: 0.9889 - loss: 0.1607 - val\_accuracy: 0.7111 - val\_loss: 0.9918  
Epoch 100/120  
23/23 ————— 6s 259ms/step - accuracy: 0.9694 - loss: 0.2016 - val\_accuracy: 0.7333 - val\_loss: 0.9014  
Epoch 101/120  
23/23 ————— 5s 211ms/step - accuracy: 0.9861 - loss: 0.1716 - val\_accuracy: 0.7556 - val\_loss: 0.9237  
Epoch 102/120  
23/23 ————— 6s 262ms/step - accuracy: 0.9806 - loss: 0.1789 - val\_accuracy: 0.7333 - val\_loss: 0.9882  
Epoch 103/120  
23/23 ————— 5s 217ms/step - accuracy: 0.9861 - loss: 0.1717 - val\_accuracy: 0.6667 - val\_loss: 1.0654  
Epoch 104/120  
23/23 ————— 5s 208ms/step - accuracy: 0.9889 - loss: 0.1717 - val\_accuracy: 0.6667 - val\_loss: 1.1621  
Epoch 105/120  
23/23 ————— 7s 276ms/step - accuracy: 0.9917 - loss: 0.1624 - val\_accuracy: 0.7111 - val\_loss: 1.0671  
Epoch 106/120  
23/23 ————— 5s 210ms/step - accuracy: 0.9889 - loss: 0.1710 - val\_accuracy: 0.7111 - val\_loss: 0.9917  
Epoch 107/120  
23/23 ————— 6s 244ms/step - accuracy: 0.9722 - loss: 0.1999 - val\_accuracy: 0.7333 - val\_loss: 0.9327  
Epoch 108/120  
23/23 ————— 10s 217ms/step - accuracy: 0.9722 - loss: 0.1862 - val\_accuracy: 0.7333 - val\_loss: 0.9588  
Epoch 109/120  
23/23 ————— 6s 279ms/step - accuracy: 0.9722 - loss: 0.1889 - val\_accuracy: 0.7556 - val\_loss: 1.0574  
Epoch 110/120  
23/23 ————— 9s 212ms/step - accuracy: 0.9889 - loss: 0.1655 - val\_accuracy: 0.7778 - val\_loss: 0.9780  
Epoch 111/120  
23/23 ————— 6s 281ms/step - accuracy: 0.9889 - loss: 0.1688 - val\_accuracy: 0.7556 - val\_loss: 0.9926  
Epoch 112/120  
23/23 ————— 5s 215ms/step - accuracy: 0.9806 - loss: 0.1808 - val\_accuracy: 0.6667 - val\_loss: 1.1123  
Epoch 113/120  
23/23 ————— 6s 243ms/step - accuracy: 0.9750 - loss: 0.1924 - val\_accuracy: 0.7778 - val\_loss: 1.2048

```

Epoch 114/120
23/23 _____ 10s 215ms/step - accuracy: 0.9861 - loss:
0.1699 - val_accuracy: 0.6889 - val_loss: 1.1632
Epoch 115/120
23/23 _____ 6s 279ms/step - accuracy: 0.9722 - loss:
0.1847 - val_accuracy: 0.7111 - val_loss: 1.1360
Epoch 116/120
23/23 _____ 5s 211ms/step - accuracy: 0.9806 - loss:
0.1771 - val_accuracy: 0.7333 - val_loss: 1.0821
Epoch 117/120
23/23 _____ 5s 213ms/step - accuracy: 0.9861 - loss:
0.1718 - val_accuracy: 0.7778 - val_loss: 1.0448
Epoch 118/120
23/23 _____ 6s 265ms/step - accuracy: 0.9889 - loss:
0.1719 - val_accuracy: 0.7111 - val_loss: 0.9869
Epoch 119/120
23/23 _____ 5s 214ms/step - accuracy: 0.9889 - loss:
0.1607 - val_accuracy: 0.7111 - val_loss: 1.1367
Epoch 120/120
23/23 _____ 6s 268ms/step - accuracy: 0.9806 - loss:
0.1862 - val_accuracy: 0.7333 - val_loss: 1.0677

```

```
# training curves
```

```
# --- efficientnet ---
```

```

eff_acc   = eff_hist_1.history["accuracy"]      +
eff_hist_2.history["accuracy"]
eff_val   = eff_hist_1.history["val_accuracy"]  +
eff_hist_2.history["val_accuracy"]
eff_loss  = eff_hist_1.history["loss"]          +
eff_hist_2.history["loss"]
eff_vloss = eff_hist_1.history["val_loss"]      +
eff_hist_2.history["val_loss"]

```

```
# --- vanilla cnn ---
```

```

cnn_acc   = cnn_hist_1.history["accuracy"]      +
cnn_hist_2.history["accuracy"]
cnn_val   = cnn_hist_1.history["val_accuracy"]  +
cnn_hist_2.history["val_accuracy"]
cnn_loss  = cnn_hist_1.history["loss"]          +
cnn_hist_2.history["loss"]
cnn_vloss = cnn_hist_1.history["val_loss"]      +
cnn_hist_2.history["val_loss"]

```

```

fig, axes = plt.subplots(2, 2, figsize=(14, 10))
fig.suptitle("Training Curves", fontsize=14, fontweight="bold")

axes[0,0].plot(eff_acc, label="Train"); axes[0,0].plot(eff_val,
label="Val", linestyle="--")
axes[0,0].axvline(x=EPOCHS_HEAD, color="gray", linestyle=":");

```



```
axes[0,0].set_title("EfficientNetB0 — Accuracy"); axes[0,0].legend();
axes[0,0].grid(True, alpha=0.4)
```

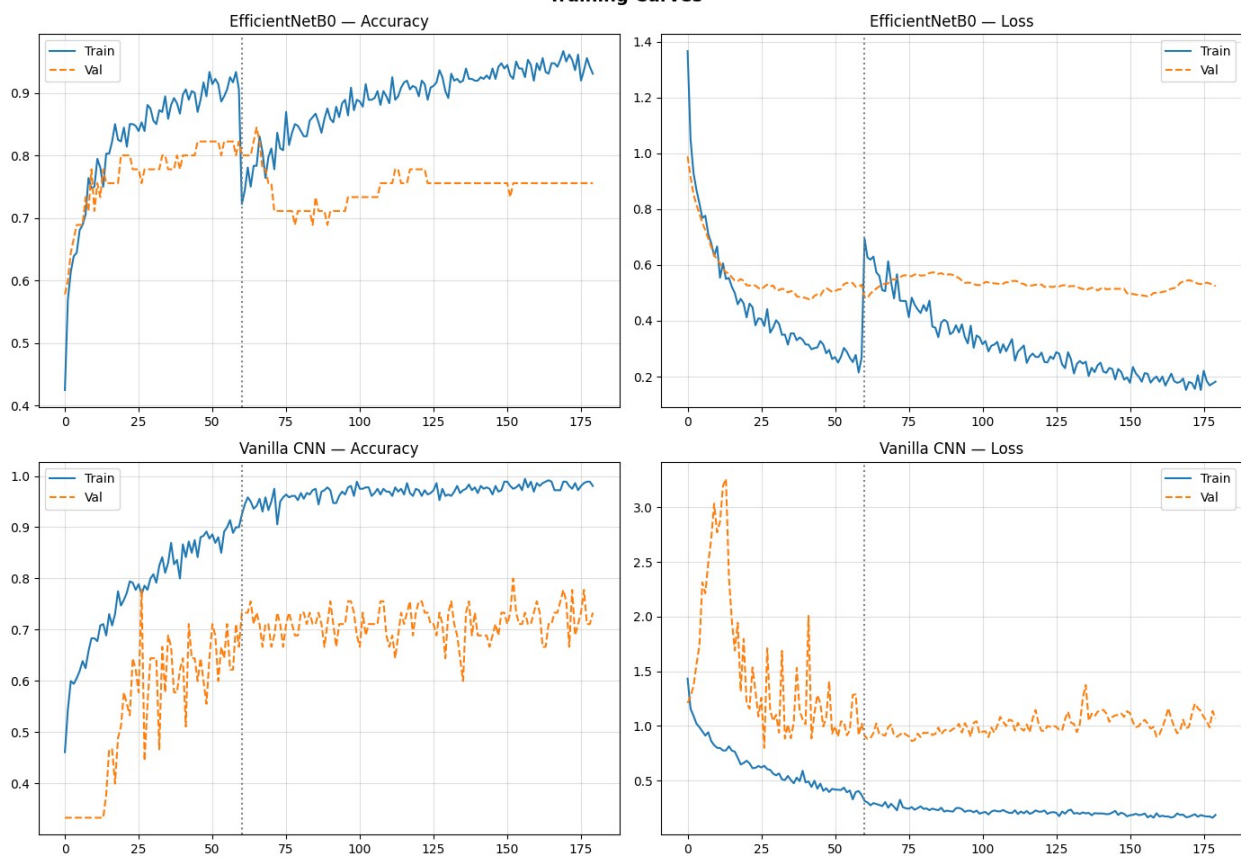
```
axes[0,1].plot(eff_loss, label="Train"); axes[0,1].plot(eff_vloss,
label="Val", linestyle="--")
axes[0,1].axvline(x=EPOCHS_HEAD, color="gray", linestyle=":");
axes[0,1].set_title("EfficientNetB0 — Loss"); axes[0,1].legend();
axes[0,1].grid(True, alpha=0.4)
```

```
axes[1,0].plot(cnn_acc, label="Train"); axes[1,0].plot(cnn_val,
label="Val", linestyle="--")
axes[1,0].axvline(x=EPOCHS_HEAD, color="gray", linestyle=":");
axes[1,0].set_title("Vanilla CNN — Accuracy"); axes[1,0].legend();
axes[1,0].grid(True, alpha=0.4)
```

```
axes[1,1].plot(cnn_loss, label="Train"); axes[1,1].plot(cnn_vloss,
label="Val", linestyle="--")
axes[1,1].axvline(x=EPOCHS_HEAD, color="gray", linestyle=":");
axes[1,1].set_title("Vanilla CNN — Loss"); axes[1,1].legend();
axes[1,1].grid(True, alpha=0.4)
```

```
plt.tight_layout(); plt.show()
```

Training Curves



```

# confusion matrices

# --- efficientnet ---
eff_probs =
eff_model.predict(tf.data.Dataset.from_tensor_slices(X_test).batch(BAT
CH_SIZE), verbose=0)
eff_preds = np.argmax(eff_probs, axis=1)

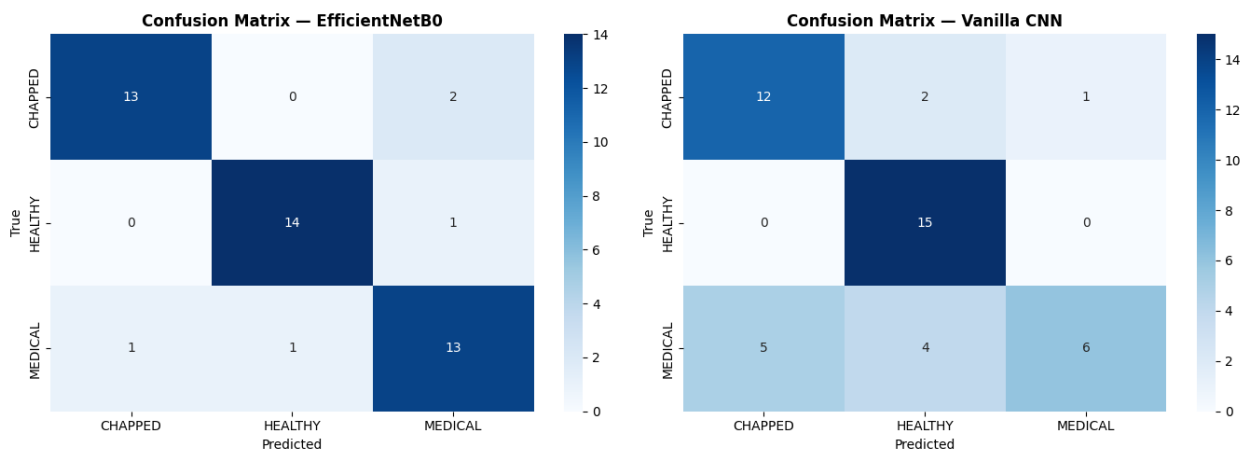
# --- vanilla cnn ---
cnn_probs =
cnn_model.predict(tf.data.Dataset.from_tensor_slices(X_test).batch(BAT
CH_SIZE), verbose=0)
cnn_preds = np.argmax(cnn_probs, axis=1)

fig, axes = plt.subplots(1, 2, figsize=(14, 5))
sns.heatmap(confusion_matrix(y_test, eff_preds), annot=True, fmt="d",
cmap="Blues",
            xticklabels=CLASSES, yticklabels=CLASSES, ax=axes[0])
axes[0].set_title("Confusion Matrix – EfficientNetB0",
fontweight="bold")
axes[0].set_xlabel("Predicted"); axes[0].set_ylabel("True")

sns.heatmap(confusion_matrix(y_test, cnn_preds), annot=True, fmt="d",
cmap="Blues",
            xticklabels=CLASSES, yticklabels=CLASSES, ax=axes[1])
axes[1].set_title("Confusion Matrix – Vanilla CNN", fontweight="bold")
axes[1].set_xlabel("Predicted"); axes[1].set_ylabel("True")

plt.tight_layout(); plt.show()

```



```

# roc curves
y_bin = label_binarize(y_test, classes=[0, 1, 2])
colors = ["#e74c3c", "#2ecc71", "#3498db"]

fig, axes = plt.subplots(1, 2, figsize=(14, 6))

```

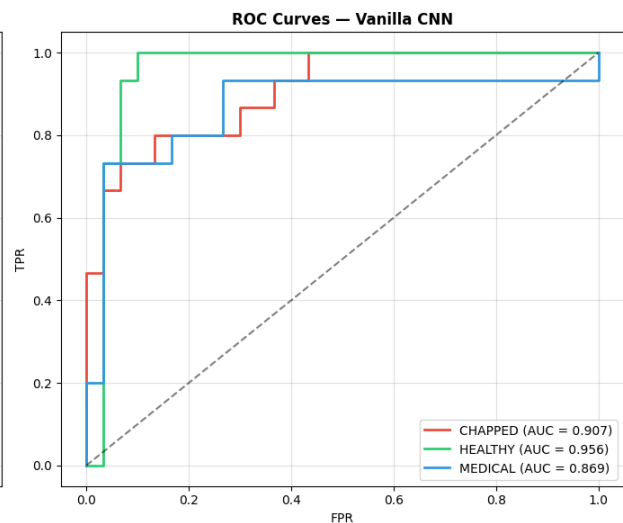
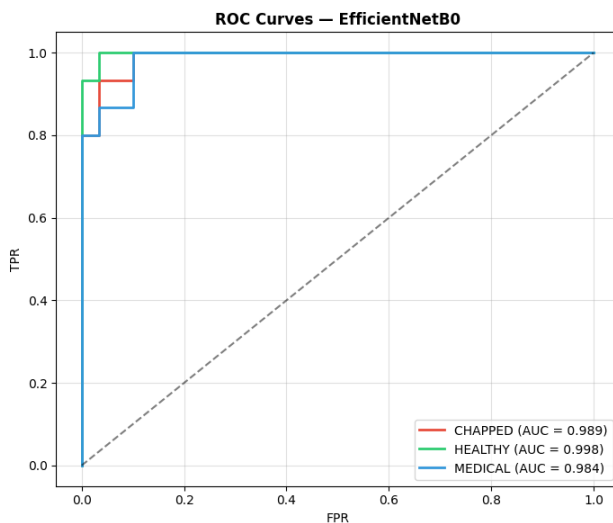
```

# --- efficientnet ---
for i, (cls, color) in enumerate(zip(CLASSES, colors)):
    fpr, tpr, _ = roc_curve(y_bin[:, i], eff_probs[:, i])
    axes[0].plot(fpr, tpr, color=color, lw=2, label=f"{cls} (AUC = {auc(fpr, tpr):.3f})")
axes[0].plot([0,1],[0,1], "k--", alpha=0.5)
axes[0].set_title("ROC Curves - EfficientNetB0", fontweight="bold")
axes[0].set_xlabel("FPR"); axes[0].set_ylabel("TPR");
axes[0].legend(); axes[0].grid(True, alpha=0.4)

# --- vanilla cnn ---
for i, (cls, color) in enumerate(zip(CLASSES, colors)):
    fpr, tpr, _ = roc_curve(y_bin[:, i], cnn_probs[:, i])
    axes[1].plot(fpr, tpr, color=color, lw=2, label=f"{cls} (AUC = {auc(fpr, tpr):.3f})")
axes[1].plot([0,1],[0,1], "k--", alpha=0.5)
axes[1].set_title("ROC Curves - Vanilla CNN", fontweight="bold")
axes[1].set_xlabel("FPR"); axes[1].set_ylabel("TPR");
axes[1].legend(); axes[1].grid(True, alpha=0.4)

plt.tight_layout(); plt.show()

```



```

# classification reports

# --- efficientnet ---
print("EfficientNetB0")
print(classification_report(y_test, eff_preds, target_names=CLASSES,
digits=4))

# --- vanilla cnn ---
print("Vanilla CNN")

```

```
print(classification_report(y_test, cnn_preds, target_names=CLASSES,
digits=4))
```

EfficientNetB0

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| CHAPPED      | 0.9286    | 0.8667 | 0.8966   | 15      |
| HEALTHY      | 0.9333    | 0.9333 | 0.9333   | 15      |
| MEDICAL      | 0.8125    | 0.8667 | 0.8387   | 15      |
| accuracy     |           |        | 0.8889   | 45      |
| macro avg    | 0.8915    | 0.8889 | 0.8895   | 45      |
| weighted avg | 0.8915    | 0.8889 | 0.8895   | 45      |

Vanilla CNN

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| CHAPPED      | 0.7059    | 0.8000 | 0.7500   | 15      |
| HEALTHY      | 0.7143    | 1.0000 | 0.8333   | 15      |
| MEDICAL      | 0.8571    | 0.4000 | 0.5455   | 15      |
| accuracy     |           |        | 0.7333   | 45      |
| macro avg    | 0.7591    | 0.7333 | 0.7096   | 45      |
| weighted avg | 0.7591    | 0.7333 | 0.7096   | 45      |

```
# comparison table
```

```
print(f"{'Metric':<15} {'EfficientNetB0':>18} {'Vanilla CNN':>14}")
print("-" * 50)
print(f"{'Accuracy':<15} {accuracy_score(y_test, eff_preds):>18.4f}
{accuracy_score(y_test, cnn_preds):>14.4f}")
print(f"{'Precision':<15} {precision_score(y_test, eff_preds,
average='weighted', zero_division=0):>18.4f} {precision_score(y_test,
cnn_preds, average='weighted', zero_division=0):>14.4f}")
print(f"{'Recall':<15} {recall_score(y_test, eff_preds,
average='weighted', zero_division=0):>18.4f} {recall_score(y_test,
cnn_preds, average='weighted', zero_division=0):>14.4f}")
print(f"{'F1-Score':<15} {f1_score(y_test, eff_preds,
average='weighted', zero_division=0):>18.4f} {f1_score(y_test,
cnn_preds, average='weighted', zero_division=0):>14.4f}")
```

| Metric    | EfficientNetB0 | Vanilla CNN |
|-----------|----------------|-------------|
| Accuracy  | 0.8889         | 0.7333      |
| Precision | 0.8915         | 0.7591      |
| Recall    | 0.8889         | 0.7333      |
| F1-Score  | 0.8895         | 0.7096      |

```
# export tflite
```

```
# --- efficientnet ---
```

```
converter = tf.lite.TFLiteConverter.from_keras_model(eff_model)
converter.optimizations = [tf.lite.Optimize.DEFAULT]
with open("/content/drive/MyDrive/efficientnet.tflite", "wb") as f:
    f.write(converter.convert())
print("Saved efficientnet.tflite")
```

```
# --- vanilla cnn ---
```

```
converter = tf.lite.TFLiteConverter.from_keras_model(cnn_model)
converter.optimizations = [tf.lite.Optimize.DEFAULT]
with open("/content/drive/MyDrive/vanilla_cnn.tflite", "wb") as f:
    f.write(converter.convert())
print("Saved vanilla_cnn.tflite")
```

Saved artifact at '/tmp/tmp0i7erfep'. The following endpoints are available:

\* Endpoint 'serve'

args\_0 (POSITIONAL\_ONLY): TensorSpec(shape=(None, 224, 224, 3), dtype=tf.float32, name='keras\_tensor\_244')

Output Type:

TensorSpec(shape=(None, 3), dtype=tf.float32, name=None)

Captures:

140425968358736: TensorSpec(shape=(1, 1, 1, 3), dtype=tf.float32, name=None)

140425968360272: TensorSpec(shape=(1, 1, 1, 3), dtype=tf.float32, name=None)

140426811695248: TensorSpec(shape=(), dtype=tf.resource, name=None)

140426811694672: TensorSpec(shape=(), dtype=tf.resource, name=None)

140426811695824: TensorSpec(shape=(), dtype=tf.resource, name=None)

140426811696592: TensorSpec(shape=(), dtype=tf.resource, name=None)

140426811697168: TensorSpec(shape=(), dtype=tf.resource, name=None)

140426811695440: TensorSpec(shape=(), dtype=tf.resource, name=None)

140426811695632: TensorSpec(shape=(), dtype=tf.resource, name=None)

140426811696976: TensorSpec(shape=(), dtype=tf.resource, name=None)

140426811697552: TensorSpec(shape=(), dtype=tf.resource, name=None)

140426811698128: TensorSpec(shape=(), dtype=tf.resource, name=None)

140426811696784: TensorSpec(shape=(), dtype=tf.resource, name=None)

140426811698896: TensorSpec(shape=(), dtype=tf.resource, name=None)

140426811697360: TensorSpec(shape=(), dtype=tf.resource, name=None)

140426811699088: TensorSpec(shape=(), dtype=tf.resource, name=None)

140426811696208: TensorSpec(shape=(), dtype=tf.resource, name=None)

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140426811697744: TensorSpec(shape=(), dtype=tf.resource, name=None)

140426811699472: TensorSpec(shape=(), dtype=tf.resource, name=None)

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140426811701008: TensorSpec(shape=(), dtype=tf.resource, name=None)

140426811696400: TensorSpec(shape=(), dtype=tf.resource, name=None)

140426811699664: TensorSpec(shape=(), dtype=tf.resource, name=None)

140426811701392: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426811700816: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426811702160: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426811699856: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426811701584: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426811702544: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426811699280: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426811700624: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426811702352: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426811702736: TensorSpec(shape=(), dtype=tf.resource, name=None)  
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140426807394768: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426811703120: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426811702928: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807394384: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807394576: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807395152: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807396304: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807396880: TensorSpec(shape=(), dtype=tf.resource, name=None)  
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140426807394960: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807396688: TensorSpec(shape=(), dtype=tf.resource, name=None)  
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140426807397072: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807398224: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807395536: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807397648: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807398032: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807396496: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807398800: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807397456: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807399568: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807399760: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807399184: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807398416: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807400144: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807400720: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807399376: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807398992: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807400528: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807401104: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807401680: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807400912: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807402064: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807399952: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807401488: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807401872: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426807400336: TensorSpec(shape=(), dtype=tf.resource, name=None)

[illegible]

140426803136080: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426803136656: TensorSpec(shape=(), dtype=tf.resource, name=None)  
140426803134928: TensorSpec(shape=(), dtype=tf.resource, name=None)  
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Saved efficientnet.tflite
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Saved artifact at '/tmp/tmpem9q9dc9'. The following endpoints are available:

\* Endpoint 'serve'

args\_0 (POSITIONAL\_ONLY): TensorSpec(shape=(None, 224, 224, 3), dtype=tf.float32, name='keras\_tensor\_251')

Output Type:

TensorSpec(shape=(None, 3), dtype=tf.float32, name=None)

Captures:

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Saved vanilla_cnn.tflite
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